

Time Series Analysis

Lecture 1: Introduction

Biqing Cai
Associate Professor

School of Economics, HUST

Spring, 2024

Outline of This Lecture

- 1 Organizational Issues
- 2 Outline of the Course
- 3 Review of Probability and Statistics
- 4 Introduction to Time Series

Organizational Issues

Instructor Information: Biqing Cai, Email:

BiqingCai@hust.edu.cn, Office: Economics Building 539

TA: Liping Wan, Email: 3081830833@qq.com, Minyi Li,
Email: 809040454@qq.com;

Office hours: Thursday 14:00-16:00 or by designation via email

Public mailbox: hust_tsanalysis@126.com (hust1234)

Textbook: Enders, W., 2015. [Applied Econometric Time Series, 4th Edition](#). Wiley.

Website: <http://www.time-series.net/>.

Organizational Issues

Reference books:

Cochrane, J.H., 2005. **Time Series for Macroeconomics and Finance**. Available at the personal website of Cochrane.

Hamilton, J.D., 1994. **Time Series Analysis**. Princeton University Press.

Tsay, R.S., 2010. **Analysis of Financial Time Series, 3rd Edition**. Wiley.

Shumway, R.H. and D.S., Stoffer, 2011. **Time Series Analysis and Its Applications (with R Examples), 3rd Edition**. Springer.

Organizational Issues

Prerequisites: Econometrics I (fundamental Probability, Statistics and Econometrics)

Software: R-language; download the free software and packages: <http://cran.r-project.org/>

Time series data: <http://robjhyndman.com/TSDL/>

Grading: There will be 4-6 homework. I will also require a small data analysis project before the end of semester. Weight of grade:

Homework: 20%;

Empirical Project: 30%;

Final Exam: 50%.

Outline of the Course

- Univariate stationary time series: AR, MA, ARMA models (Chapter 1-2 of the textbook).

In 1950s-1970s, macroeconomic time series analysis was dominated by the simultaneous equations models (L.R. Klein won the Noble prize 1980). The problem is forecasting performance is not so good. Box, G.E. and G.M., Jenkins, 1976. **Time series analysis: forecasting and control, rev. ed.** San Francisco: Holde-Day. ARIMA modelling.

- Volatility models (conditional heteroskedasticity): ARCH, GARCH models (Chapter 3 of the textbook).

Engle (1982) proposed ARCH model and won the Nobel prize 2003.

Outline of the Course

- Multivariate time series: VAR models (Chapter 5 of the textbook).

Sims (1980) proposed VAR model and won the Nobel prize 2011.

- Nonstationary time series: unit root, cointegration models (Chapter 4 and 6).

Engle and Granger (1987) proposed cointegration models and won the Nobel prize 2003.

- Topics on nonlinear time series (if time permits).

Review of Probability and Statistics

There are two fundamental axioms in modern econometrics:

- Axiom A: Any economy can be viewed as stochastic process governed by some probability laws;
- Axiom B: Economic Phenomenon, as often summarized in form of data, can be viewed as a realization of the stochastic data generating process (DGP).

⇒ we need to have “statistical thinking” in economic analysis.

Review of Probability and Statistics

A good reference book: **Lecture Notes on Probability and Statistics Theory for Economists** (by Prof. Yongmiao Hong).

- **Probability Space.** A probability space is a triple $(S, \mathbb{B}, P)$ where:
 - S is the sample space corresponding to the outcomes of the underlying random experiments.
 - $\mathbb{B}$ is the σ -field of subsets of S . These subsets are called events.
 - $P : \mathbb{B} \rightarrow [0, 1]$ is a probability measure.

Example: coin throwing. $S = \{-1, 1\}$, $\mathbb{B} = \{\emptyset, S, \{1\}, \{-1\}\}$ and $P(x = -1) = P(x = 1) = 1/2$.

Review of Probability and Statistics

- Probability densities and distributions (pmf/pdf, CDF): discrete and continuous random variables. The CDF of a random variable X is defined as:

$$F_X(x) = P(X \leq x) \text{ for all } x \in \mathbb{R}.$$

- Expectation and population moments. Suppose X is a random variable with pmf or pdf $f_X(x)$. Then the expectation of a measurable function $g(X)$ is defined as

$$E(g(X)) = \int_{-\infty}^{\infty} g(x) dF_X(x).$$

As a special case, when $g(X) = X$, it is the first moment (mean).

Review of Probability and Statistics

- Joint distributions and independence. The joint CDF of X and Y is defined as follows:

$$F_{XY}(x, y) = P(X \leq x, Y \leq y) = P(X \leq x \cap Y \leq y),$$

for any pair $(x, y) \in \mathbb{R}^2$. The random variables X and Y are independent if and only if

$$F_{XY}(x, y) = F_X(x)F_Y(y),$$

for all $(x, y) \in \mathbb{R}^2$.

- Law of iterated expectation. On a probability space with two sub σ -algebras $\mathbb{G}_1 \subseteq \mathbb{G}_2 \subseteq \mathbb{B}$, for a random variable X , we have

$$E[E[X | \mathbb{G}_2] | \mathbb{G}_1] = E[X | \mathbb{G}_1].$$

Review of Probability and Statistics

Intuition: The smaller sub σ -algebra dominates!

Example 1: Consider the regression model $Y = X\beta + \varepsilon$. The assumption $E(\varepsilon | X) = 0$ is stronger than $E(\varepsilon) = 0$ or $E(X\varepsilon) = 0$ because the first condition implies the second condition.

Example 2: In macroeconomics or finance, for the Overlapping-generations model (OLG), we often use the property $E_t(X) = E_t(E_{t+1}(X))$, where E_t denotes the expectation conditional on the information (σ -algebra) up to time t .

Review of Probability and Statistics

- Statistics. Let $X^n = \{X_1, X_2, \dots, X_n\}$ be a random sample of size n from a population. A statistic $T_n = T(X^n) = T(X_1, \dots, X_n)$ is a real-valued or vector-valued function of the random sample.

Example: Sample moments. $\frac{\sum_{i=1}^n X_i}{n}$ is the sample mean.

No any unknown parameter is involved!

- Convergence in probability. A sequence of random variables $\{Z_n\}$ converges in probability to a random variable Z if for any small $\epsilon > 0$,

$$P[|Z_n - Z| > \epsilon] \rightarrow 0 \text{ as } n \rightarrow \infty.$$

It can be written $Z_n \rightarrow_p Z$.

Example: (Weak) Consistency of the parameter estimation

$$\widehat{\mu} \rightarrow_p \mu.$$

Unbiasedness and Consistency

Unbiasedness: Letting $\widehat{\beta}$ be an estimator for unknown parameter β , it is an unbiased estimator if $E(\widehat{\beta}) = \beta$.

Remark: Unbiasedness is a finite sample property while consistency is a large sample (asymptotic) property.

Example: We have n i.i.d. observations $\{x_1, \dots, x_n\}$ with mean μ variance σ^2 . How to estimate μ ? Three possibilities: (1) x_1 ; (2) $\widehat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i$ (sample mean); (3) $\widehat{\mu} + x_1/n$.

Review of Probability and Statistics

- (Weak) Law of Large Numbers (LLN). Let $X_1, X_2, \dots$ be i.i.d. random variables with $E(X_i) = \mu$ and $\text{var}(X_i) < \infty$. Define $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Then we have $\bar{X}_n \rightarrow_p \mu$.

Remark: This is the simplest LLN. **How about not i.i.d.?** Two directions: independent but not identical; not independent (time series case); Multivariate case. White (2001, Chapter 2-5) contains more rigorous treatments.

- Convergence in distribution. Let $\{Z_n\}$, $n = 1, 2, \dots$ be a sequence of random variables with a sequence of corresponding cdf's $F_n(z)$, $n = 1, 2, \dots$ and let Z be a random variable cdf $F(z)$. Then Z_n converge in distribution to Z as $n \rightarrow \infty$ if the cdf of $F_n(z)$ converges to $F(z)$ at all continuity points.

It can be written as $Z_n \rightarrow_d Z$.

Review of Probability and Statistics

- Central Limit Theorem (CLT). [**Lindeberg-Levy CLT:**] Let $X_1, X_2, \dots$ be a sequence of i.i.d. random variables with mean μ and variance $0 < \sigma^2 < \infty$. Then we have $\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \rightarrow_d N(0, 1)$ as $n \rightarrow \infty$.

It is possible to have CLT under more general conditions and multivariate case.

- Slutsky Theorem. Suppose $X_n \rightarrow_d X$ and $c_n \rightarrow_p c$, a constant. Then we have
 - $X_n \pm c_n \rightarrow_d X \pm c$
 - $X_n c_n \rightarrow_d Xc$
 - $\frac{X_n}{c_n} \rightarrow_d \frac{X}{c}$ for $c \neq 0$

Example: Replacing σ in the CLT by their consistent estimators.

CLT for Dependent Data

For time series data (dependent process), we have following version of central limit theorem.

Theorem 1

Assume $E x_t = 0$ and $\sigma^2 = \gamma_0 + 2 \sum_{j=1}^{\infty} \gamma_j$ is positive. Then

$$\sum_{t=1}^T x_t / \sqrt{T} \rightarrow_d N(0, \sigma^2)$$

if one of the following two conditions holds: (1) $E |x_t|^\delta < \infty$ and $\sum_{j \geq 1} \alpha(j)^{1-2/\delta} < \infty$ for some constant $\delta > 2$; (2) $P(|x_t| < c) = 1$ for some constant c and $\sum_{j \geq 1} \alpha(j) < \infty$, where $\alpha(j)$ are the strong mixing coefficients.

Proof: See the proof of Theorem 2.21 in Fan and Yao (2003).

Review of Probability and Statistics

- Continuous mapping theorem. If $c_n \rightarrow_p c$, where c is a constant, g is a continuous function, then $g(c_n) \rightarrow_p g(c)$; If $X_n \rightarrow_d X$, g is a continuous function, then $g(X_n) \rightarrow_d g(X)$.

Example: If $x_n \rightarrow_d N(0, 1)$, we have $x_n^2 \rightarrow_d \chi^2(1)$.

Review of Probability and Statistics

Stochastic Order of Magnitude

$Z_n = o_P(n^\alpha)$ if $Z_n/n^\alpha \rightarrow_p 0$.

$Z_n = O_P(n^\alpha)$ if for given $\delta > 0$, there exists a constant $\Delta(\delta) < \infty$ and a finite integer $N(\delta)$ such that $P(|Z_n/n^\alpha| > \Delta) < \delta$ for all $n > N$.

Remark 1: When $Z_n = o_P(n^\alpha)$, then $Z_n = O_P(n^\alpha)$. A random variable is $O_P(1)$. $o_P(1)$ is asymptotically negligible.

Remark 2: If $X_n = O_P(n^\lambda)$ and $Y_n = O_P(n^\mu)$, then $X_n Y_n = O_P(n^{\lambda+\mu})$ and $X_n + Y_n = O_P(n^\kappa)$ where $\kappa = \max(\lambda, \mu)$; If $X_n = o_P(n^\lambda)$ and $Y_n = o_P(n^\mu)$, then $X_n Y_n = o_P(n^{\lambda+\mu})$ and $X_n + Y_n = o_P(n^\kappa)$ where $\kappa = \max(\lambda, \mu)$; If $X_n = O_P(n^\lambda)$ and $Y_n = o_P(n^\mu)$, then $X_n Y_n = o_P(n^{\lambda+\mu})$ and $X_n + Y_n = O_P(n^\kappa)$ where

Review of Probability and Statistics

- Ordinary Least Squares (OLS) and Maximum Likelihood Estimation (MLE). A simple example: suppose we have observations $x_1, \dots, x_n$ sampling from i.i.d. random variable $X_1, \dots, X_n$ with mean μ . How to estimate μ ?
 - OLS. Regress X on a constant and minimize $\sum_{i=1}^n (x_i - \mu)^2$, then we get a estimator $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i$, which is the sample mean.
 - MLE. We further assume that the true distribution is normal and the variance is σ^2 . Then the likelihood function

$$f_{X_1, \dots, X_n}(\mu, \sigma) = \prod_{i=1}^n f_{X_i}(\mu, \sigma),$$

with $f_{X_i}(\mu, \sigma)$ being the normal density with mean μ and variance σ^2 . We can show that the MLE of μ is $\frac{1}{n} \sum_{i=1}^n x_i$.

Types of Data

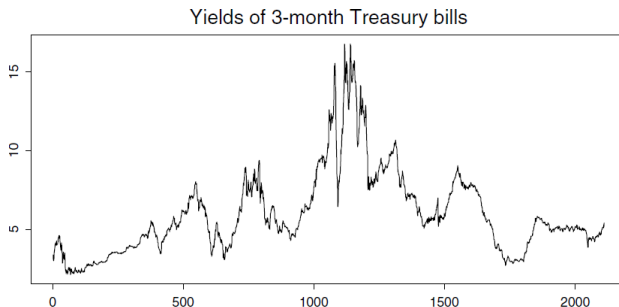
Basically, there are 3 types of data.

- Cross sectional data. E.g., productivity of different firms.
- Time series data. E.g., GDP growth rate of China from 1980 to 2015.
- Panel data. E.g., GDP growth rate of the provinces in China from 1980 to 2015.

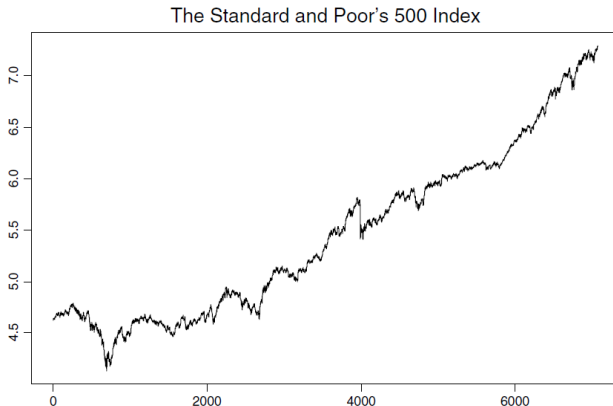
New developments: Big data: text data; network data; functional data (e.g. interval data)

What is Time Series?

Figure illustrations.



This is the 3-month T-bill rate from July 17, 1959 to December 31, 1999.



This is the logarithm of Standard and Poor's 500 index from January 3, 1972 to December 31, 1999.

Roughly speaking, a time series is a collection of observations indexed by the date of each observation.

Formal Definition

- Stochastic process. A stochastic process is a family of random variable $\{X_t, t \in T\}$ in a probability space $(S, \mathbb{B}, P)$.

Example: Random Walk process. Let $\{e_t, t = 1, 2, \dots\}$ be i.i.d. process. Then the process $S_t = S_0 + \sum_{j=1}^t e_j$ for $t = 1, 2, \dots$ is a random walk process.

- Realization of stochastic process. The functions $\{X(s), s \in S\}$ on T are known as the realizations of the or sample-paths of the process $\{X_t, t \in T\}$.

Time series. We frequently use the term **time series** the data and the process of which it is a realization. Let X_t be a stochastic process, $X_t(s)$ is a realization of the stochastic process.

Formal Definition

Example:, we consider the following **time series model** (AR(1) and random walk):

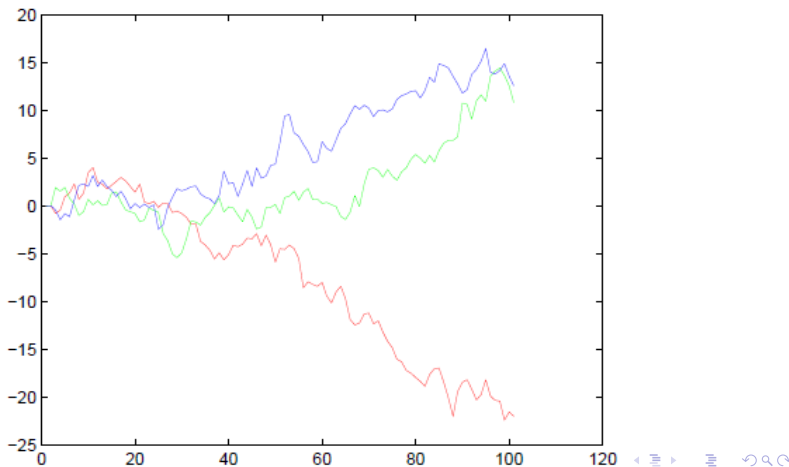
$$x_t = x_{t-1} + e_t,$$

where $x_0 = 0$ and $e_t \sim i.i.d.N(0, 1)$, $t = 1, \dots, 100$.

Time series mentions both the model and the realizations (data).

Formal Definition

The following figure gives 3 realizations (sample paths) of the model.



Objectives of Time Series Analysis

- Description of data.

Example: Classical decomposition: $X_t = T_t + S_t + C_t + e_t$
where T_t : Trend term; S_t : seasonal component; C_t : cyclical component and e_t : irregular component.

- Forecasting.

Example: Predict sales, GDP growth, interest rate, unemployment rate, stock prices.

Objectives of Time Series Analysis

- Testing economic theories.

Example: The Unbiased Forward Rate (UFR) hypothesis asserts the following relationship:

$$s_{t+1} = f_t + \varepsilon_{t+1},$$

where s_t is a particular foreign currency on the spot market against dollar (the benchmark) at time t , f_t is the price of the currency for delivery one period into the future against dollar. Then testing UFR corresponds to testing $\alpha_0 = 0$ and $\alpha_1 = 1$ in the following model:

$$s_{t+1} = \alpha_0 + \alpha_1 f_t + \varepsilon_{t+1}.$$

Time Series Analysis

Lecture 2: Difference Equations

Biqing Cai
Associate Professor

School of Economics, HUST

Spring, 2024

Outline of This Lecture

- 1 Introduction
- 2 Linear Difference Equation and Its Solution
- 3 The Cobweb Model
- 4 Solving Homogeneous Difference Equations
- 5 Finding Particular Solutions
- 6 Homework

Two Terminologies

I will first introduce two notations that will be used in this lecture.

- Difference operator Δ . $\Delta y_t \equiv y_t - y_{t-1}$.

Similarly, we can define second difference $\Delta^2 y_t \equiv \Delta(\Delta y_t) = \Delta(y_t - y_{t-1}) = (y_t - y_{t-1}) - (y_{t-1} - y_{t-2}) = y_t - 2y_{t-1} + y_{t-2}$.

- Lag operator L . $L^i y_t = y_{t-i}$. Properties 1-6 in the textbook (p.40). We can see that $\Delta y_t = y_t - Ly_t$.

Why Difference Equations?

Think about the classic decomposition:

$$y_t = T_t + S_t + I_t,$$

and let $T_t = 1 + 0.1t$, $S_t = 1.6 \sin(t\pi/6)$ and $I_t = 0.7I_{t-1} + \varepsilon_t$ with ε_t being a pure random disturbance. Each of these three equations is a type of **difference equation**. In its most general form, difference equation expresses the value of a variable as a function of its own lagged values, time and other variables.

Time series econometrics is concerned with the estimation of difference equations containing stochastic components.

Linear Difference Equation

The form of an n -th order linear difference equation with constant coefficients is given as follows:

$$y_t = a_0 + \sum_{i=1}^n a_i y_{t-i} + x_t, \quad (1)$$

where x_t is called the **forcing process**. x_t can be function of t , current or lagged values of other variables, or random disturbances (e.g. $x_t = \sum_{i=0}^{\infty} \beta_i \varepsilon_{t-i}$).

It can be written in terms of the difference operator

$$\Delta y_t = a_0 + \gamma y_{t-1} + \sum_{i=2}^n a_i y_{t-i} + x_t,$$

where $\gamma = a_1 - 1$.

Solution to Difference Equation

A solution to a difference equation expresses the value of y_t as a function of the elements of $\{x_t\}$ and t (and possibly some given values of y_t sequence called initial condition). Analogy to calculus.

Example 1: $\Delta y_t = 2$. A solution is $y_t = 2t + c$ where c is an arbitrary constant. We can verify this.

Example 2: $I_t = 0.7I_{t-1} + \varepsilon_t$ (the irregular component). A solution is $I_t = \sum_{i=0}^{\infty} 0.7^i \varepsilon_{t-i}$. We can verify this.

The above solutions are simply postulated? **Are there methods to obtain the solutions?**

Solution by Iteration

Consider the first difference equation

$$y_t = a_0 + a_1 y_{t-1} + \varepsilon_t, \quad (2)$$

given the value of y_0 .

Forward solution gives:

$$y_1 = a_0 + a_1 y_0 + \varepsilon_1$$

$$y_2 = a_0(1 + a_1) + a_1^2 y_0 + \varepsilon_2 + a_1 \varepsilon_1$$

and for all $t > 0$, repeated iteration yields (Note there is typo in the textbook p.10)

$$y_t = a_0 \sum_{i=0}^{t-1} a_1^i + a_1^t y_0 + \sum_{i=0}^{t-1} a_1^i \varepsilon_{t-i}$$

Solution by Iteration

Another way of iteration: **Backward** iteration.

$$\begin{aligned}y_t &= a_0 + a_1(a_0 + a_1 y_{t-1} + \varepsilon_{t-1}) + \varepsilon_t \\&= a_0(1 + a_1) + a_1 \varepsilon_{t-1} + \varepsilon_t + a_1^2(a_0 + a_1 y_{t-2} + \varepsilon_{t-2}) \\&= \dots\end{aligned}$$

Continuing iteration back to period 0, we have

$$y_t = a_0 \sum_{i=0}^{t-1} a_1^i + a_1^t y_0 + \sum_{i=0}^{t-1} a_1^i \varepsilon_{t-i} \quad (3)$$

Solution by Iteration: Unknown Initial Condition

Suppose we are not given the initial condition for y_0 , for y_0 we still have

$$y_0 = a_0 + a_1 y_{-1} + \varepsilon_0.$$

Thus continuing to iterate back for 1 period, for (3) we have

$$y_t = a_0 \sum_{i=0}^t a_1^i + \sum_{i=0}^t a_1^i \varepsilon_{t-i} + a_1^{t+1} y_{-1} \quad (4)$$

and continuing to iterate back for m periods, we obtain

$$y_t = a_0 \sum_{i=0}^{t+m} a_1^i + \sum_{i=0}^{t+m} a_1^i \varepsilon_{t-i} + a_1^{t+m+1} y_{-m-1} \quad (5)$$

Remark 1: If $|a_1| < 1$, we have from (5)

$$y_t = \frac{a_0}{1 - a_1} + \sum_{i=0}^{\infty} a_1^i \varepsilon_{t-i} \quad (6)$$

We can verify this by substituting (6) into (2).

Remark 2: We can also show that for arbitrary value of A , a solution to (2) is given by

$$y_t = Aa_1^t + \frac{a_0}{1 - a_1} + \sum_{i=0}^{\infty} a_1^i \varepsilon_{t-i} \quad (7)$$

We can verify this by substituting (7) into (2).

Remark 3: When y_0 is given, we can identify A by noticing

$$y_0 = A + \frac{a_0}{1 - a_1} + \sum_{i=0}^{\infty} a_1^i \varepsilon_{t-i}$$

So that

$$y_t = \left[y_0 - \frac{a_0}{1 - a_1} \right] a_1^t + \frac{a_0}{1 - a_1} + \sum_{i=0}^{t-1} a_1^i \varepsilon_{t-i} \quad (8)$$

We can verify that (3) and (8) are identical.

Remark 4: If the condition $|a_1| \geq 1$, we can not get the result (6) (look at (5)), however the result (3) still holds. As a special case, when $a_1 = 1$,

$$y_t = a_0 t + \sum_{i=1}^t \varepsilon_i + y_0$$

An Alternative Solution

Solution by iteration breaks down in higher order equations (algebra complexity quickly overwhelms any reasonable attempts to find solutions).

To study a solution methodology, we consider the **homogeneous** portion of (2)

$$y_t = a_1 y_{t-1} \tag{9}$$

The solution to the homogeneous equation is called the **homogeneous solution** (denoted by y_t^h). An obvious trivial solution to (8) is

$$y_t = y_{t-1} = \cdots = 0$$

An Alternative Solution

An easy to verify solution to (9) is

$$y_t = Aa_1^t,$$

with A an arbitrary constant.

Remark: If $|a_1| < 1$, $a_1^t \rightarrow 0$ as $t \rightarrow \infty$; if $|a_1| > 1$, the homogeneous solution is not stable (if $a_1 > 1$, the solution approaches ∞ as $t \rightarrow \infty$; if $a_1 < -1$, the solution oscillates explosively); if $a_1 = 1$, arbitrary constant A ; if $a_1 = -1$, meta-stable.

An Alternative Solution

We have confirmed that (6) is a solution to (2). It is called a **particular solution** to the difference equation because the solution may not be *unique*.

The **general solution** to a difference equation is defined to be a particular equation plus all homogeneous solutions.

An Alternative Solution

The results for the first order case are directly applicable to the n -th order equation. The difference is (1) more difficult to find the particular solution and (2) n (rather than 1) distinct homogenous solutions.

The solution methodology entails following 4 steps:

- 1 form the homogeneous equation and find all n homogeneous solutions
- 2 find a particular solution
- 3 obtain the general solution as the sum of the particular solution and a linear combination of all homogeneous solutions
- 4 eliminate the arbitrary constants by imposing the initial conditions on the solution

An Alternative Solution

To illustrate the solution methodology, we consider the model

$$y_t = 0.9y_{t-1} - 0.2y_{t-2} + 3 \quad (10)$$

- ① form the homogeneous equation $y_t - 0.9y_{t-1} + 0.2y_{t-2} = 0$.
Two homogenous solutions are $y_{1t}^h = 0.5^t$ and $y_{2t}^h = 0.4^t$
(general method to find these two solutions will be discussed later).
- ② find a particular solution $y_t = 10$.
- ③ combine the particular solution and linear combination of homogeneous solutions and yield $y_t = A_1 0.5^t + A_2 0.4^t + 10$.
- ④ suppose $y_0 = 13$ and $y_1 = 11.3$, then A_1 and A_2 satisfy
 $13 = A_1 + A_2 + 10$ and $11.3 = 0.5A_1 + 0.4A_2 + 10 \Rightarrow A_1 = 1$
and $A_2 = 2$.

Generalizing the Method

To show the method is applicable to higher order equations, consider the homogeneous part of (1)

$$y_t = \sum_{i=1}^n a_i y_{t-i}$$

We will shown later that, there are n homogeneous solutions. Then we can use the 4 steps.

The Cobweb Model

As an illustration of the methodology, we consider a stochastic version of the Cobweb model

$$d_t = a - \gamma p_t \quad (11)$$

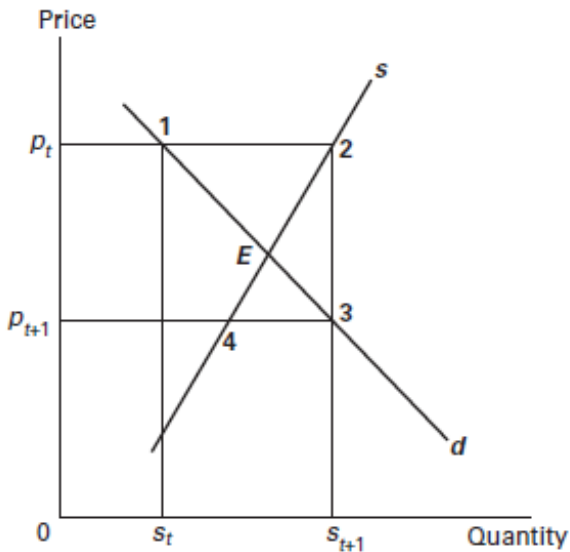
$$s_t = b + \beta p_t^* + \varepsilon_t \quad (12)$$

$$s_t = d_t \quad (13)$$

Where d_t is the demand of wheat in period t , s_t is the supply of wheat in period t , p_t is the market price of wheat in period t , p_t^* is the price that farmers expect to prevail in t and ε_t is a zero mean stochastic supply shock. And parameters a , b , γ and β are all positive with $a > b$.

Furthermore, it is assumed that the farmers use last year's price as the market price, i.e. $p_t^* = p_{t-1}$. The model can be shown in the following figure.

The Cobweb Model



The Cobweb Model

From the figure, we can see that the market will always converges to the long-run equilibrium point. **Is it always the case?**
We have

$$b + \beta p_{t-1} + \varepsilon_t = a - \gamma p_t$$

and accordingly

$$p_t = (-\beta/\gamma)p_{t-1} + (a - b)/\gamma - \varepsilon_t/\gamma \quad (14)$$

The solution to (14), we follow the four steps as described above.

- Form the homogenous equation $p_t = (-\beta/\gamma)p_{t-1}$. We can show that the homogeneous solution is

$$p_t^h = A(-\beta/\gamma)^t$$

The Cobweb Model

- By iterating back and if $\beta/\gamma < 1$, we have

$$p_t^p = \frac{a-b}{\gamma+\beta} - \frac{1}{\gamma} \sum_0^{\infty} (-\beta/\gamma)^i \varepsilon_{t-i}.$$

If $\beta/\gamma \geq 1$, we can not get the result and it is necessary to impose an initial condition.

- The general solution is:

$$p_t = \frac{a-b}{\gamma+\beta} - \frac{1}{\gamma} \sum_0^{\infty} (-\beta/\gamma)^i \varepsilon_{t-i} + A(-\beta/\gamma)^t$$

The Cobweb Model

- If we know y_0 , A can be eliminated by noticing

$$p_0 = \frac{a-b}{\gamma+\beta} - \frac{1}{\gamma} \sum_0^{\infty} (-\beta/\gamma)^i \varepsilon_{-i} + A(-\beta/\gamma)^0$$

Thus the value of A is given by

$$A = p_0 - \frac{a-b}{\gamma+\beta} + \frac{1}{\gamma} \sum_0^{\infty} (-\beta/\gamma)^i \varepsilon_{-i}$$

So that the solution is

$$p_t = \frac{a-b}{\gamma+\beta} - \frac{1}{\gamma} \sum_0^{\infty} (-\beta/\gamma)^i \varepsilon_{t-i} + (-\beta/\gamma)^t \left(p_0 - \frac{a-b}{\gamma+\beta} + \frac{1}{\gamma} \sum_0^{\infty} (-\beta/\gamma)^i \varepsilon_{-i} \right)$$

which can be simplified as

$$p_t = \frac{a-b}{\gamma+\beta} - \frac{1}{\gamma} \sum_0^{t-1} (-\beta/\gamma)^i \varepsilon_{t-i} + (-\beta/\gamma)^t \left(p_0 - \frac{a-b}{\gamma+\beta} \right) \quad (15)$$

The Cobweb Model

Remark 1: If there are not stochastic shock, i.e. all the values of ε_t are zero. If the system start with $p_0 = (a - b)/(\gamma + \beta)$ (the long-run equilibrium), looking at (15), we have $p_t = (a - b)/(\gamma + \beta)$, i.e., the system stays at the long-run equilibrium.

Remark 2: Suppose the process begins at a price below long-run equilibrium: $p_0 < (a - b)/(\gamma + \beta)$, then from (15),

$$p_1 = (a - b)/(\gamma + \beta) + [p_0 - (a - b)/(\gamma + \beta)](-\beta/\gamma) > (a - b)/(\gamma + \beta)$$

and

$$p_2 = (a - b)/(\gamma + \beta) + [p_0 - (a - b)/(\gamma + \beta)](-\beta/\gamma)^2 < (a - b)/(\gamma + \beta)$$

If $\beta/\gamma < 1$, oscillation will diminish and the price converges to long-run equilibrium; if $\beta/\gamma > 1$, the oscillations will be explosive.

The Cobweb Model

Remark 3: Consider the effect of the stochastic shock. The partial derivative of p_t with respect to ε_t (contemporaneous effect)

$$\frac{\partial p_t}{\partial \varepsilon_t} = \frac{-1}{\gamma} \quad (16)$$

(16) is called the **impact multiplier**.

The effect of the supply shock in t persist into future periods, we have the **one-period multiplier**

$$\frac{\partial p_{t+1}}{\partial \varepsilon_t} = \frac{-1}{\gamma}(-\beta/\gamma) = \beta/\gamma^2$$

and furthermore

$$\frac{\partial p_{t+n}}{\partial \varepsilon_t} = \frac{-1}{\gamma}(-\beta/\gamma)^n.$$

The time path of all such multipliers is called **impulse response function**.

The Cobweb Model

Remark 4: The three components in (15) have economic interpretations.

- The first component is the long-run equilibrium. If the stability condition is met, p_t sequence tends to the equilibrium
- The second component catches the short-run price adjustment to the supply shocks
- The third component is an initial effect. if $\beta/\gamma < 1$, the effect diminishes over time

Solving Second order Homogeneous Difference Equation

Consider the second order equation

$$y_t - a_1 y_{t-1} - a_2 y_{t-2} = 0 \quad (17)$$

Given the findings for the first order case, we suspect that the solution takes the form of $y_t^h = A\alpha^t$. Substitute it into (17), we have

$$A\alpha^t - a_1 A\alpha^{t-1} - a_2 A\alpha^{t-2} = 0$$

and accordingly

$$\alpha^2 - a_1 \alpha - a_2 = 0$$

Solving Second order Homogeneous Difference Equation

Solving this quadratic equation (**characteristic equation**) yields two **characteristic roots**

$$\begin{aligned}\alpha_1, \alpha_2 &= \frac{a_1 \pm \sqrt{a_1^2 + 4a_2}}{2} \\ &= (a_1 \pm \sqrt{d})/2,\end{aligned}$$

where d is the discriminant $a_1^2 + 4a_2$.

Remark: Each of the two characteristic roots solves (17), but the solution is not unique. It is easy to see that a complete homogenous solution takes the form $y_t^h = A_1\alpha_1^t + A_2\alpha_2^t$ for arbitrary A_1 and A_2 .

Solving Second order Homogeneous Difference Equation

We can further discuss the solution based on three possible case of the discriminant d .

Case 1: $d > 0$, then there are two distinct real characteristic roots. Hence

$$y_t^h = A_1 \alpha_1^t + A_2 \alpha_2^t$$

Case 2: $d = 0$, then $\alpha_1 = \alpha_2 = a_1/2$. However, there is another homogeneous solution $t(a_1/2)^t$ (we can verify this). A complete homogeneous solution is

$$y_t^h = A_1 (a_1/2)^t + A_2 t (a_1/2)^t$$

Solving Second order Homogeneous Difference Equation

Case 3: $d < 0$, then the characteristic roots are imaginary:

$$\alpha_1 = (a_1 + i\sqrt{-d})/2 \text{ and } \alpha_2 = (a_1 - i\sqrt{-d})/2.$$

$$y_t^h = A_1 \alpha_1^t + A_2 \alpha_2^t,$$

where $i = \sqrt{-1}$. The homogenous solution is

$$y_t^h = \beta_1 r^t \cos(\theta t + \beta_2)$$

where β_1 and β_2 are arbitrary constants, $r = (-a_2)^{1/2}$ and θ is chosen to be

$$\cos(\theta) = a_1/[2(-a_2)^{1/2}]$$

Stability Condition

Case 1: The largest root less than 1 and the smallest root greater than -1

Case 2: The stability condition is $|a_1| < 2$

Case 3: $r < 1$

In summary, stability requires all characteristic roots lie within the unit circle.

Higher Order Systems

The n -th polynomial will yield n solutions for α , denoted by $\alpha_1, \dots, \alpha_n$. Then $A_1\alpha_1^t + \dots + A_n\alpha_n^t$ is also a solution. And the constants will be eliminated by imposing n initial conditions.

Facts about Stability: a necessary condition is $\sum_{i=1}^n a_i < 1$; a sufficient condition is $\sum_{i=1}^n |a_i| < 1$; at least one characteristic root equals to 1 if $\sum_{i=1}^n a_i = 1$.

Particular Solution for Deterministic Process

Finding a particular solution is often a matter of ingenuity. The appropriate technique depends heavily on the x_t process. We separate into 3 cases:

Case 1: $x_t = 0$. Then the constant $c = a_0/(1 - a_1 - \dots - a_n)$ is a solution. If $1 - a_1 - \dots - a_n = 0$ (unit root case), it seems that a linear trend is possible and we try $y_t^p = ct$, then

$$ct = a_0 + a_1 c(t-1) + \dots + a_n c(t-n)$$

and thus

$$c = a_0/(a_1 + 2a_2 + \dots + na_n)$$

If the solution ct fails, try $ct^2, \dots, ct^n$.

Particular Solution for Deterministic Process

Case 2: $x_t = b(d)^{rt}$, the exponential case. Then

$$y_t = a_0 + a_1 y_{t-1} + b d^{rt}$$

Suppose the solution is $y_t^p = c_0 + c_1 d^{rt}$ and make substitutions, we get

$$c_0 = a_0 / (1 - a_1)$$

$$c_1 = b d^{rt} / (d^r - a_1)$$

If $a_1 = 1$, try $c_0 = ct$; if $a_1 = d^r$, try $c_1 = tb$.

Particular Solution for Deterministic Process

Case 3: $x_t = bt^d$, the deterministic time trend with d a positive integer. Then

$$y_t = a_0 + \sum_{i=1}^n a_i y_{t-i} + bt^d$$

Suppose the solution is $y_t^p = c_0 + c_1 t + \cdots + c_d t^d$ and make substitutions.

Particular Solution for Stochastic Process

There are two useful methods for finding particular solutions when there are stochastic components in $\{y_t\}$ process.

- The **method of undetermined coefficients**. Idea: a particular solution to linear difference equation is linear and the solution can depend only on time, a constant and the element of $\{x_t\}$ (positing a solution called **challenge solution**). Thus it is possible to know the exact form of the solution even when the coefficients of the solution are unknown. Case 2 and Case 3 for the deterministic case.
- The **lag operator approach**. Utilizing the lag operator, ϕ often more convenient than the method of undetermined coefficients.

The Method of Undetermined Coefficients

Consider the first order equation

$$y_t = a_0 + a_1 y_{t-1} + \varepsilon_t$$

We posit the challenge solution

$$y_t = b_0 + b_1 t + \sum_{i=0}^{\infty} \alpha_i \varepsilon_{t-i},$$

where b_0 , b_1 and α_i are the coefficients to be determined.

Substitute the solution into the original equation, we have

$$b_0 + b_1 t + \alpha_0 \varepsilon_t + \cdots = a_0 + a_1 [b_0 + b_1 (t-1) + \alpha_0 \varepsilon_{t-1} + \cdots] + \varepsilon_t$$

The Method of Undetermined Coefficients

Collecting like terms, each of the conditions must hold:

$$\alpha_0 - 1 = 0$$

$$\alpha_1 - a_1 \alpha_0 = 0$$

.

.

.

$$b_0 - a_0 - a_1 b_0 + a_1 b_1 = 0$$

$$b_1 - a_1 b_1 = 0$$

If $a_1 \neq 1$,

$$y_t = \frac{a_0}{1 - a_1} + \sum_{i=0}^{\infty} a_1^i \varepsilon_{t-i}$$

Same as the one derived by iteration method.

The Method of Undetermined Coefficients

Remark 1: Combining the result of homogeneous solution, we have the general solution

$$y_t = \frac{a_0}{1 - a_1} + \sum_{i=0}^{\infty} a_1^i \varepsilon_{t-i} + Aa_1^t$$

And if there is an initial condition y_0 , it follows that

$$y_t = \frac{a_0}{1 - a_1} + \sum_{i=0}^{t-1} a_1^i \varepsilon_{t-i} + a_1^t [y_0 - a_0/(1 - a_1)]$$

Remark 2: If $a_1 = 1$, b_0 can be arbitrary constant and $b_1 = a_0$. If we have initial value y_0 , the solution is

$$y_t = y_0 + a_0 t + \sum_{i=1}^t \varepsilon_i$$

Remark 3: Similar procedure for higher order system.

Lag Operator Approach

The n -th order equation can be rewritten as

$$(1 - a_1 L - \cdots - a_p L^p)y_t \equiv A(L)y_t = a_0 + \varepsilon_t$$

So that

$$y_t = (a_0 + \varepsilon_t)/A(L)$$

Remark: The stability condition is that the **inverse characteristic equation** $1 - a_1 L - \cdots - a_n L^n$ has characteristic roots **outside** the unit circle.

Lag Operator Approach

To illustrate the approach, consider the second order case and we assume that we know

$1 - a_1L - a_2L^2 = (1 - b_1L)(1 - b_2L)$, then

$$y_t = (a_0 + \varepsilon_t) / [(1 - b_1L)(1 - b_2L)]$$

If both b_1 and b_2 are less than 1 in absolute value, we can apply property 5 of lag operator in the textbook

$$y_t = \frac{a_0 / (1 - b_1) + \sum_{i=0}^{\infty} b_1^i \varepsilon_{t-i}}{1 - b_2L}$$

We can reapply property 5 to obtain the particular solution.

Remark: The general model is $A(L)y_t = a_0 + B(L)\varepsilon_t$ we can have particular solution $y_t = a_0/A(L) + B(L)\varepsilon_t/A(L)$.

Homework

Exercise 1, 3, 7.

Note that the homework need to be submitted one week later.

Those exceed the deadline will not be accepted.

The SIR Model

The Susceptible-Infective-Removal (SIR) model (Kermack and McKendrick (1927)) is a fundamental epidemiology model.

$$\begin{aligned}\frac{dS_t}{dt} &= -\beta I(t) \frac{S(t)}{N} \\ \frac{dI(t)}{dt} &= \beta I(t) \frac{S(t)}{N} - \gamma I(t) \\ \frac{dR(t)}{dt} &= \gamma I(t)\end{aligned}$$

where β and γ are parameters. Important information can be obtained from the model, e.g., β is related to the so-called reproduction number ($R_0 = \beta D$, where D is the duration). Many generalizations.

Time Series Analysis

Lecture 3: Stationary Time Series Models

Biqing Cai

Associate Professor

School of Economics, HUST

Spring, 2024

Outline of This Lecture

- 1 Introduction
- 2 Stationarity
- 3 The ACF and PACF
- 4 Estimation, Testing and Model Selection
- 5 Forecast
- 6 A Model of Interest Rate Spread
- 7 Seasonality
- 8 Parameter Instability and Structure Change
- 9 Summary and Conclusions
- 10 Homework

ARMA(p,q) Model

This chapter is mainly concerned about ARMA(p,q) model taking the following form:

$$y_t = a_0 + \sum_{i=1}^p a_i y_{t-i} + \sum_{i=0}^q \beta_i \varepsilon_{t-i} \quad (1)$$

and we follow the convention of normalizing units so that $\beta_0 = 1$.

Remark 1: $\sum_{i=1}^p a_i y_{t-i}$ is the autoregressive part and if $q = 0$, it is an AR(p) process; $\sum_{i=0}^q \beta_i \varepsilon_{t-i}$ is the moving average part and if $p = 0$, the process is MA(q) process.

Remark 2: If one of the characteristic roots is equal to 1, the $\{y_t\}$ is an **integrated** process and the model is called ARIMA model.

ARMA Model

Remark 3: We can solve for y_t in terms of the $\{\varepsilon_t\}$ sequence using the lag operator approach discussed in Lecture 2. We have

$$\left(1 - \sum_{i=1}^p a_i L^i\right) y_t = a_0 + \sum_{i=0}^q \beta_i \varepsilon_{t-i}$$

so that a particular solution is

$$y_t = \left(a_0 + \sum_{i=0}^q \beta_i \varepsilon_{t-i} \right) / \left(1 - \sum_{i=1}^p a_i L^i \right)$$

The expansion will yield a $MA(\infty)$ process.

Remark 4: The process $\{\varepsilon_t\}$ is the **innovation** to the $ARMA(p,q)$ process and is the **building block** of the process. In this chapter, it is assumed that $\{\varepsilon_t\}$ is a white noise process. In the following, we discuss three related concepts.

I.I.D., M.D.S. and White Noise

The i.i.d. process. A process $\{\varepsilon_t\}$ is an i.i.d. process if the sequence $\varepsilon_1, \varepsilon_2, \dots$ is i.i.d..

The martingale difference sequence (m.d.s.). A process $\{\varepsilon_t\}$ on the probability space $(S, \mathbb{B}, P)$ is a m.d.s. if $E(\varepsilon_t | F_{t-1}) = 0$ a.s. where F_{t-1} is the information (σ -algebra) up to time $t - 1$.

The white noise process. A process $\{\varepsilon_t\}$ is a white noise process if for any time period t $E(\varepsilon_t) = 0$; $E(\varepsilon_t^2) = \sigma^2$; $E(\varepsilon_t \varepsilon_{t-s}) = E(\varepsilon_{t-j} \varepsilon_{t-j-s}) = 0$ for all j and all $s \neq 0$ (p.49 $s \neq 0$ needs to be included).

I.I.D., M.D.S. and White Noise

The i.i.d. assumption is usually adopted in nonlinear time series analysis when probability structure of the model is to be studied. The white noise assumption is used in the chapter and it suffices for us to study the ARMA type model. The m.d.s. assumption is more related to economic theory.

Example 1: The efficient market hypothesis (EMH). Denote r_t to be the return of an asset, under efficient market, we have

$$E(r_t - \mu \mid F_{t-1}) = 0,$$

where $\mu = E(r_t)$. Intuitively, the efficient market hypothesis argues that the asset return is not predictable. According to the definition of F_{t-1} , we have weak-form efficiency (the information of the price), semi-strong-form efficiency (public information) and strong-form efficiency (all information, public and private).

I.I.D., M.D.S. and White Noise

Example 2: In macroeconomic models and asset pricing theory, after solving the dynamic optimization problem, we often have

$$p_t = E(m_{t+1}x_{t+1} | F_t),$$

where P_t is the asset price at time t , x_{t+1} is the asset payoff at time $t + 1$ and m_{t+1} is the stochastic discount factor. The equation can be written as

$$E(m_{t+1}x_{t+1} - p_t | F_t) = 0$$

i.e., $\varepsilon_{t+1} \equiv m_{t+1}x_{t+1} - p_t$ is a m.d.s.

Remark: The above equation implies infinitely many moment conditions.

I.I.D., M.D.S. and White Noise

The relationship of the three processes is summarized as follows: when $E(\varepsilon_t^2) = \sigma^2$ for all t ,

$$l.i.d. \text{ (with mean zero)} \Rightarrow m.d.s \Rightarrow \text{white noise}$$

Remark: The proof of the first argument follows from independence and mean zero; the proof of the second argument follows from the law of iteration.

M.D.S. But Not I.I.D.

Consider the ARCH(1) process

$$X_t = \sigma_t e_t$$

$$\sigma_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2$$

$$e_t \sim i.i.d.(0, 1)$$

We can show that $\{X_t\}$ is a m.d.s. because $E(X_t | F_{t-1}) = 0$, but it is clear that $E(X_t^2 | F_{t-1}) = \sigma_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2$ which suggests that it is not i.i.d. process.

White Noise But Not M.D.S.

Consider the nonlinear MA process

$$\begin{aligned}X_t &= \alpha e_{t-1} e_{t-2} + e_t \\e_t &\sim i.i.d.(0, 1)\end{aligned}$$

It is easy to see that (1) $E(X_t) = 0$ (2) $E(X_t^2) = 1 + \alpha^2$ and (3) $E(X_t X_{t-s}) = 0$ when $s \neq 0$. So that it is a white noise process. But we have $E(X_t | F_{t-1}) = \alpha e_{t-1} e_{t-2} \neq 0$

课外阅读

洪永淼，2002。金融计量的新进发展。经济学（季刊），第一卷第二期，249-268.

Stationarity

In time series analysis, stationarity plays an important role. Intuitively, for statistical inference, we need to have more and more information as sample size increases. Stationarity guarantees information increases regarding certain aspects of the time series (i.e. the moments).

Consider 4 machines, and let y_{it} represent the hourly output of machine i at time t , the mean can be calculated as

$$\bar{y}_t = \frac{1}{4} \sum_{i=1}^4 y_{it}$$

$\bar{y}_t$ is called **ensemble average** whose calculation requires knowledge of multiple time series.

Stationarity

Typically, we have only **one** set of realizations of particular series. So that we have the following **time average**

$$\bar{y}_t = \frac{1}{T} \sum_{t=1}^T y_{1t}$$

If $\{y_t\}$ is a stationary time series, the mean, variance and autocorrelations can usually be well approximated by sufficiently long **time averages** based on a single set of realizations.

Stationarity

In using the time average approximation, it would be assumed that mean was the same for each period. Formally, the stochastic process is assumed to be **covariance stationary (weakly stationary, second-order stationary, wide-sense stationary)**.

I.e., the stochastic process have finite mean and variance and for all t , $t - s$ and j

$$E(y_t) = E(y_{t-s}) = \mu$$

$$E[(y_t - \mu)^2] = E[(y_{t-s} - \mu)^2] = \sigma^2$$

$$E[(y_t - \mu)(y_{t-s} - \mu)] = E[(y_{t-j} - \mu)(y_{t-s-j} - \mu)] = \gamma_s$$

Remark: Another often used terminology: **autocorrelation** between y_t and y_{t-s} is $\rho_s = \gamma_s / \gamma_0$.

Stationarity

Example 1: It is easy to see that the white noise process is weakly stationary.

Example 2: Let ε_t be a white noise process, then the moving average process $y_t = \varepsilon_t + \beta\varepsilon_{t-1}$ is weakly stationary.

Example 3: Let ε_t be a white noise process, then the trend stationary process $y_t = \beta t + \varepsilon_t$ is not weakly stationary because the mean is not a constant.

Strict Stationarity

There is another type of stationarity called **strict stationarity**. The stochastic process $\{y_t\}$ is strictly stationary if the joint distribution of $(y_{t_1}, y_{t_2}, \dots, y_{t_k})$ is the same as that of $(y_{t_1+h}, y_{t_2+h}, \dots, y_{t_k+h})$.

Remark: Strict stationarity does not implied weak stationarity, nor does weak stationarity imply strict stationarity. When the second moment of the process exists, strict stationarity implies weak stationarity.

Stationarity for an AR(1) process

Consider the AR(1) process

$$y_t = a_0 + a_1 y_{t-1} + \varepsilon_t$$

where $\{\varepsilon_t\}$ is a white noise process. And suppose the process starts at period 0 with value y_0 , then we have

$$y_t = a_0 \sum_{i=0}^{t-1} a_1^i + a_1^t y_0 + \sum_{i=0}^{t-1} a_1^i \varepsilon_{t-i}$$

So that

$$E(y_t) = a_0 \sum_{i=0}^{t-1} a_1^i + a_1^t y_0$$

and

$$E(y_{t+s}) = a_0 \sum_{i=0}^{t+s-1} a_1^i + a_1^{t+s} y_0$$

Stationarity for an AR(1) process

It is easy to see that y_t is not stationary. However, if $|a_1| < 1$, as $t \rightarrow \infty$

$$\lim_{t \rightarrow \infty} y_t = \frac{a_0}{1 - a_1} + \sum_{i=0}^{\infty} a_1^i \varepsilon_{t-i}$$

Taking expectation on this quantity, we have for sufficiently large enough t ,

$$E(y_t) = \frac{a_0}{1 - a_1} \equiv \mu$$

$$\begin{aligned} E(y_t - \mu)^2 &= E[(\varepsilon_t + a_1 \varepsilon_{t-1} + a_1^2 \varepsilon_{t-2} + \cdots)^2] \\ &= \sigma^2 / (1 - a_1^2) \end{aligned}$$

$$\begin{aligned} E[(y_t - \mu)(y_{t-s} - \mu)] &= E[\varepsilon_t + a_1 \varepsilon_{t-1} + \cdots][\varepsilon_{t-s} + a_1 \varepsilon_{t-s-1} + \cdots] \\ &= \sigma^2 (a_1^s) [1 + a_1^2 + \cdots] = \sigma^2 (a_1^s) / (1 - a_1^2) \end{aligned}$$

Stationarity for an AR(1) process

Remark: If we use the limiting value, then $\{y_t\}$ is stationary. For any given value y_0 and $|a_1| < 1$, t must be sufficiently large. In other words, the process needs to start from infinitely long time ago.

Now we consider the case without initial condition. The solution for y_t is

$$y_t = \frac{a_0}{1 - a_1} + \sum_{i=0}^{\infty} a_1^i \varepsilon_{t-i} + Aa_1^t$$

Remark: We can see that it is stationary **iff** $Aa_1^t = 0$. Stationarity requires (1) The homogenous solution is zero; (2) the characteristic root a_1 less than 1 in absolute value.

Stationarity for an ARMA(p,q) Model

For a general ARMA(p,q) model, stationarity also requires the homogenous solution to be zeros, now we are going to study the requirements for the particular solution.

We start to study an ARMA(2,1) model

$$y_t = a_1 y_{t-1} + a_2 y_{t-2} + \varepsilon_t + \beta_1 \varepsilon_{t-1} \quad (2)$$

Using the method of undetermined coefficient, we write the challenge solution

$$y_t = \sum_{i=0}^{\infty} c_i \varepsilon_{t-i}$$

Stationarity for an ARMA(p,q) Model

Substitute the challenge solution into the original equation and match the coefficients, we have

$$c_0 = 1$$

$$c_1 = a_1 c_0 + \beta_1$$

$$c_i = a_1 c_{i-1} + a_2 c_{i-2} \text{ for all } i \geq 2$$

Remark: If the characteristic roots of (2) are within the unit circle, the $\{c_i\}$ constitute a (geometrically) convergent sequence.

Stationarity for an ARMA(p,q) Model

We are now to verify that the ARMA(2,1) model is stationary characteristic roots of (2) are within the unit circle. It is easy to see that for all t and s

$$E(y_t) = E(y_{t-s}) = 0$$

$$\begin{aligned} E(y_t^2) &= E[(c_0\varepsilon_t + c_1\varepsilon_{t-1} + \cdots)^2] \\ &= \sigma^2 \sum_{i=0}^{\infty} c_i^2 \end{aligned}$$

And the covariance between y_t and y_{t-s} is

$$\text{cov}(y_t, y_{t-s}) = \sigma^2(c_s + c_{s+1}c_1 + c_{s+2}c_2 + \cdots)$$

We can see that the three quantities are independent of t , so that the process is covariance stationary.

Stationarity for an ARMA(p,q) Model

The previous result can be generalized to general ARMA(p,q) models. Consider the MA(∞) process

$$x_t = \sum_{i=0}^{\infty} \beta_i \varepsilon_{t-i} \quad (3)$$

For model (3), we have

$$E(x_t) = E(\varepsilon_t + \beta_1 \varepsilon_{t-1} + \cdots) = 0$$

$$E(x_t^2) = E[(\varepsilon_t + \beta_1 \varepsilon_{t-1} + \cdots)^2] = \sigma^2 \sum_{i=0}^{\infty} \beta_i^2$$

and

$$\begin{aligned} E(x_t x_{t-s}) &= E[(\varepsilon_t + \beta_1 \varepsilon_{t-1} + \cdots)(\varepsilon_{t-s} + \beta_1 \varepsilon_{t-s-1} + \cdots)] \\ &= \sigma^2 (\beta_s + \beta_1 \beta_{s+1} + \beta_2 \beta_{s+2} + \cdots) \end{aligned}$$

Stationarity for an ARMA(p,q) Model

We can summarize that the **necessary and sufficient** conditions for any MA process to be stationary is that

$$\sum_{i=0}^{\infty} \beta_i^2$$

and for all $s > 0$

$$(\beta_s + \beta_1\beta_{s+1} + \beta_2\beta_{s+2} + \cdots)$$

to be finite.

Remark 1: In fact, using the Cauchy-Schwarz inequality, we have $(\sum_{k=0}^{\infty} \beta_k \beta_{s+k})^2 \leq (\sum_{k=0}^{\infty} \beta_k^2)(\sum_{k=0}^{\infty} \beta_{s+k}^2) < \infty$ (the textbook does not mention about this).

Remark 2: All the **finite** order MA processes are stationary.

Stationarity for an ARMA(p,q) Model

We now consider ARMA(p,q) model and start with considering the AR(p) model

$$y_t = a_0 + \sum_{i=1}^p a_i y_{t-i} + \varepsilon_t$$

If the characteristic roots of the homogeneous equation all lie in the unit circle, it is possible to write the particular solution

$$y_t = a_0 / (1 - \sum_{i=1}^p a_i) + \sum_{i=0}^{\infty} c_i \varepsilon_{t-i}$$

where c_i are undetermined coefficients.

Remark: When the characteristic roots of the homogeneous equation all lie in the unit circle, $\{c_i\}$ sequence is **geometrically** convergent which implies $\sum_{i=0}^{\infty} c_i^2$ is finite.

Stationarity for an ARMA(p,q) Model

We have

$$E(y_t) = a_0 / (1 - \sum_{i=1}^p a_i)$$

$$\text{var}(y_t) = \sigma^2 \sum_{i=0}^{\infty} c_i^2$$

and

$$\text{cov}(y_t, y_{t-s}) = \sigma^2 (c_s + c_1 c_{s+1} + \dots)$$

Thus we conclude that the process is covariance stationary.

Remark: For ARMA(p,q) process, when the characteristic roots of the homogeneous equation all lie in the unit circle, we can show that it is covariance stationary. **Only the roots of the autoregressive portion determine the stationarity of ARMA(p,q) process!**

The Autocorrelation Function for AR(1) Process

The autocovariances and autocorrelations serve as useful tool in the Box-Jenkins (1976) approach to identifying and estimating time series models. For the AR(1) model (**when it is stationary, i.e., $|a_1| < 1$**), we have

$$\begin{aligned}\gamma_0 &= \sigma^2 / (1 - a_1^2) \\ \gamma_s &= \sigma^2 (a_1^s) / (1 - a_1^2)\end{aligned}$$

And forming the autocorrelations by dividing each γ_s by γ_0 , we find that

$$\begin{aligned}\rho_0 &= 1 \\ \rho_s &= a_1^s\end{aligned}$$

The plot of ρ_s against s -called the **autocorrelation function (ACF)** or **correlogram** converges to 0 geometrically.

The Autocorrelation Function for AR(2) Process

We consider the AR(2) model

$$y_t = a_1 y_{t-1} + a_2 y_{t-2} + \varepsilon_t$$

If the roots of $1 - a_1 L - a_2 L^2 = 0$ are outside the unit circle, the process is stationary.

Now we apply the technique of **Yule-Walker** equations. Multiply the equation by y_{t-s} for $s = 0, 1, 2, \dots$ and take expectation

$$\begin{aligned} E y_t y_t &= a_1 E y_{t-1} y_t + a_2 E y_{t-2} y_t + E \varepsilon_t y_t \\ E y_t y_{t-1} &= a_1 E y_{t-1} y_{t-1} + a_2 E y_{t-2} y_{t-1} + E \varepsilon_t y_{t-1} \\ &\vdots \\ E y_t y_{t-s} &= a_1 E y_{t-1} y_{t-s} + a_2 E y_{t-2} y_{t-s} + E \varepsilon_t y_{t-s} \end{aligned}$$

The Autocorrelation Function for AR(2) Process

From the above equations, we have

$$\begin{aligned}\gamma_0 &= a_1\gamma_1 + a_2\gamma_2 + \sigma^2 \\ \gamma_1 &= a_1\gamma_0 + a_2\gamma_1 + 0 \\ &\vdots \\ \gamma_s &= a_1\gamma_{s-1} + a_2\gamma_{s-2} + 0\end{aligned}$$

So that

$$\begin{aligned}\rho_1 &= a_1\rho_0 + a_2\rho_1 \\ \rho_s &= a_1\rho_{s-1} + a_2\rho_{s-2}\end{aligned}$$

By $\rho_0 = 1$, we derive that $\rho_1 = a_1/(1 - a_2)$ and then derive ρ_s for all $s \geq 2$.

The Autocorrelation Function for MA(1) Process

Consider the MA(1) process

$$y_t = \varepsilon_t + \beta\varepsilon_{t-1}$$

We can also apply the Yule-Walker equations to derive

$$\gamma_0 = E y_t y_t = E[(\varepsilon_t + \beta\varepsilon_{t-1})(\varepsilon_t + \beta\varepsilon_{t-1})] = (1 + \beta^2)\sigma^2$$

$$\gamma_1 = E[(\varepsilon_t + \beta\varepsilon_{t-1})(\varepsilon_{t-1} + \beta\varepsilon_{t-2})] = \beta\sigma^2$$

and

$$\gamma_s = E[(\varepsilon_t + \beta\varepsilon_{t-1})(\varepsilon_{t-s} + \beta\varepsilon_{t-s-1})] = 0 \text{ for all } s > 1$$

Hence, $\rho_0 = 1$, $\rho_1 = \beta/(1 + \beta^2)$ and $\rho_s = 0$ for all $s > 1$.

The Autocorrelation Function for ARMA(1,1) Process

Consider the ARMA(1,1) process

$$y_t = a_1 y_{t-1} + \varepsilon_t + \beta \varepsilon_{t-1}$$

We can also apply the Yule-Walker equations to derive

$$\gamma_0 = a_1 \gamma_1 + \sigma^2 + \beta_1(a_1 + \beta_1)\sigma^2$$

$$\gamma_1 = a_1 \gamma_0 + \beta_1 \sigma^2$$

$$\gamma_2 = a_1 \gamma_1$$

$$\vdots$$

$$\gamma_s = a_1 \gamma_{s-1}$$

The Autocorrelation Function for ARMA(1,1) Process

Thus we have

$$\begin{aligned}\gamma_0 &= \frac{1 + \beta_1^2 + 2a_1\beta_1}{1 - a_1^2} \sigma^2 \\ \gamma_1 &= \frac{(1 + a_1\beta_1)(a_1 + \beta_1)}{1 - a_1^2} \sigma^2 \\ &\vdots \\ \gamma_s &= a_1 \gamma_{s-1}\end{aligned}$$

And then we can obtain ρ_s accordingly.

The Partial Autocorrelation Function

In AR(1) process, when the lag is larger than 2, the correlation does not disappear ($\rho_2 = \rho_1^2$). The **Partial autocorrelation** between y_t and y_{t-s} eliminates the effects of the intervening value y_{t-1} through y_{t-s+1} .

As such, in an AR(1) model, the partial autocorrelation between y_t and y_{t-2} is 0.

Construction of Partial Autocorrelation Function

The steps to construct the partial autocorrelation function is as follows:

- 1 Demean and construct $\{y_t^*\}$ sequence with $y_t^* = y_t - \mu$
- 2 Form the first order autocorrelation $y_t^* = \phi_{11}y_{t-1}^* + e_t$. We can see that ϕ_{11} is both the autocorrelation and partial autocorrelation
- 3 Control for (netting out) the effect of y_{t-1} on the correlation between y_t and y_{t-2} by constructing the second order autocorrelation $y_t^* = \phi_{21}y_{t-1}^* + \phi_{22}y_{t-2}^* + e_t$. ϕ_{22} the the partial autocorrelation of y_t and y_{t-2}
- 4 Repeating the procedure for all additional lags s yield the **partial autocorrelation function (PACF)**

The Partial Autocorrelation Function

Due to the construction, we have

$$\begin{aligned}\phi_{11} &= \rho_1 \\ \phi_{22} &= \frac{\rho_2 - \rho_1^2}{1 - \rho_1^2} \\ \phi_{ss} &= \frac{\rho_s - \sum_{j=1}^{s-1} \phi_{s-1,j} \rho_{s-j}}{1 - \sum_{j=1}^{s-1} \phi_{s-1,j} \rho_j}, \quad s = 3, 4, 5, \dots\end{aligned}$$

where $\phi_{s,j} = \phi_{s-1,j} - \phi_{s,s} \phi_{s-1,s-j}$ $j = 1, 2, \dots, s-1$.

Remark: We know that for a MA(q) model, the ACF will be 0 for all $s > q$ and for AR(p) model, the PACF will be 0 for all $s > p$. This is a useful feature to identification of MA and AR model respectively.

Properties of the ACF and PACF

Table 2.1 Properties of the ACF and PACF

Process	ACF	PACF
White noise	All $\rho_s = 0$ ($s \neq 0$)	All $\phi_{ss} = 0$
AR(1): $a_1 > 0$	Direct geometric decay: $\rho_s = a_1^s$	$\phi_{11} = \rho_1$; $\phi_{ss} = 0$ for $s \geq 2$
AR(1): $a_1 < 0$	Oscillating decay: $\rho_s = a_1^s$	$\phi_{11} = \rho_1$; $\phi_{ss} = 0$ for $s \geq 2$
AR(p)	Decays toward zero. Coefficients may oscillate.	Spikes through lag p . All $\phi_{ss} = 0$ for $s > p$
MA(1): $\beta > 0$	Positive spike at lag 1. $\rho_s = 0$ for $s \geq 2$	Oscillating decay: $\phi_{11} > 0$
MA(1): $\beta < 0$	Negative spike at lag 1. $\rho_s = 0$ for $s \geq 2$	Geometric decay: $\phi_{11} < 0$
ARMA(1, 1): $a_1 > 0$	Geometric decay beginning after lag 1. Sign $\rho_1 = \text{sign}(a_1 + \beta)$	Oscillating decay after lag 1. $\phi_{11} = \rho_1$
ARMA(1, 1): $a_1 < 0$	Oscillating decay beginning after lag 1. Sign $\rho_1 = \text{sign}(a_1 + \beta)$	Geometric decay beginning after lag 1. $\phi_{11} = \rho_1$ and $\text{sign}(\phi_{ss}) = \text{sign}(\phi_{11})$
ARMA(p, q)	Decay (either direct or oscillatory) beginning after lag q	Decay (either direct or oscillatory) beginning after lag p

Figure Illustration of the ACF and PACF

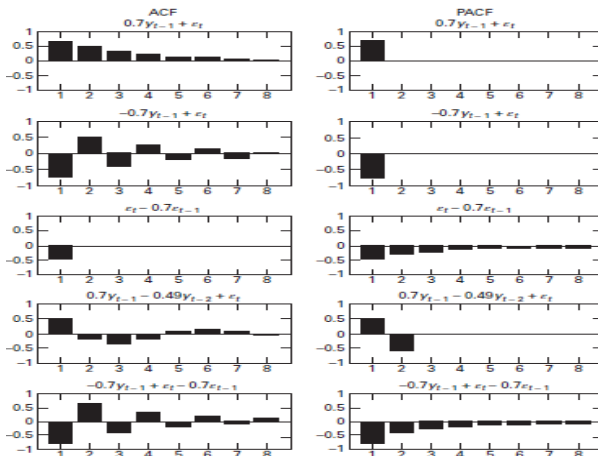


FIGURE 2.2 Theoretical ACF and PACF Patterns

Sample Autocorrelation of Stationary Time Series

In practice, we have no idea about the true DGP. But given stationarity, we can use sample mean, variance and autocorrelation to estimate their population counterparts where

$$\begin{aligned}\bar{y} &= \frac{1}{T} \sum_{t=1}^T y_t \\ \widehat{\sigma}^2 &= \frac{1}{T} \sum_{t=1}^T (y_t - \bar{y})^2 \\ r_s &= \frac{\sum_{t=s+1}^T (y_t - \bar{y})(y_{t-s} - \bar{y})}{\sum_{t=1}^T (y_t - \bar{y})^2}\end{aligned}$$

Remark: Under some regularity conditions, it can be shown that the quantities are consistent estimators for their population counterparts. The asymptotic normality can also be established.

Sample Partial Autocorrelation Function

Similarly, we can also compute the sample partial autocorrelation function. Consider the following regression

$$y_t = b_0 + b_1 y_{t-1} + \cdots + b_p y_{t-p} + e_t$$

And the estimators (by OLS) $\widehat{b}_0, \dots, \widehat{b}_p$ are the sample PACF.

Remark: For the MA(s-1) model, then $\text{Var}(r_s) = 1/T$ for $s = 1$ and $\text{Var}(r_s) = \frac{1}{T}(1 + 2 \sum_{j=1}^{s-1} \rho_j^2)$ for $s > 1$; and for the AR(p) model, $\widehat{b}_{p+i}$ $i \geq 1$ converges in distribution to a normal variable with variance approximately $1/T$. So that confidence intervals can be constructed to judge the order of MA and AR model respectively (by seeing whether the estimators are larger than $\pm 1.96 \sqrt{\text{var}(r_s)}$ or $\pm 1.96 \sqrt{\text{var}(\widehat{b}_s)}$ respectively).

Figure Illustration of Sample ACF and PACF

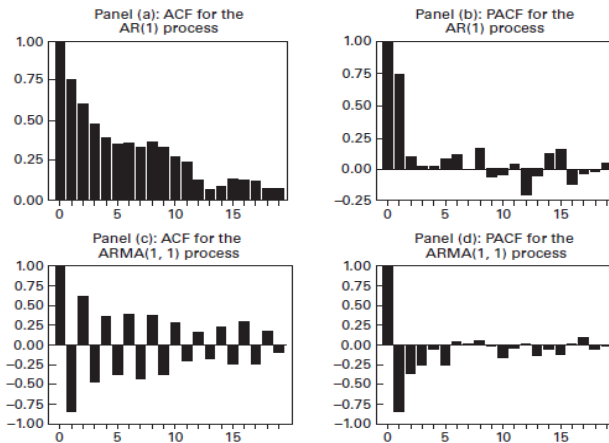


FIGURE 2.3 ACF and PACF for Two Simulated Processes

Parameter Estimation in ARMA models

For an AR(p) model, we can use OLS method to estimate the parameters, i.e.

$$(\widehat{b}_0, \dots, \widehat{b}_p) = \arg \min_{b_0, \dots, b_p} \sum_{t=p+1}^T (y_t - b_0 - \sum_{i=1}^p b_i y_{t-i})^2$$

However, for the ARMA models or MA models, since $\{\varepsilon_t\}$ are not observable, we can not use OLS and MLE is to be used.

Remark: Even when MLE is used, the methodology can be quite complicated. So that in what follows, we just illustrate how to use MLE in i.i.d. case can show how to use it in MA(1) model.

The Procedure of MLE Method

The general procedure to obtain MLE estimators is as follows

- 1 Derive the likelihood function and taking logarithm
- 2 Derive the first order condition and obtain the parameter estimators by solving the first order conditions
- 3 Check the second order condition, i.e. examine whether the **Hessian matrix** is negative definite

To illustrate the procedure, we consider the following example.

MLE in Multiple Regression

In multiple linear regression $Y = X\beta + \varepsilon$, we assume that

$$\varepsilon|X \sim \mathcal{N}(0, \sigma^2 I).$$

So that the conditional likelihood function for y_i is

$$f_{Y_i|X} = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(Y_i - x_i'\beta)^2}{2\sigma^2}\right).$$

Then the likelihood function can be written as

$$\begin{aligned} L(\beta, \sigma^2) &= \prod_{i=1}^T f_{Y_i|X} = (2\pi\sigma^2)^{-\frac{n}{2}} \exp\left(-\frac{\varepsilon'\varepsilon}{2\sigma^2}\right) \\ &= (2\pi\sigma^2)^{-\frac{n}{2}} \exp\left(-\frac{(Y - X\beta)'(Y - X\beta)}{2\sigma^2}\right). \end{aligned}$$

MLE in Multiple Regression

Taking logarithm, we have

$$\log L(\beta, \sigma^2) = -\frac{T}{2} \log(2\pi) - \frac{T}{2} \log \sigma^2 - \frac{1}{2\sigma^2} (Y - X\beta)'(Y - X\beta)$$

The MLE estimator $(\widehat{\beta}, \widehat{\sigma}^2)$ satisfies

$$(\widehat{\beta}, \widehat{\sigma}^2) = \arg \max_{\beta, \sigma^2} \log L(\beta, \sigma^2).$$

Then we have the first order condition

$$\begin{cases} \frac{\partial \log L}{\partial \beta} \mid (\beta = \widehat{\beta}, \sigma^2 = \widehat{\sigma}^2) = -\frac{1}{2\widehat{\sigma}^2} (-2X'Y + 2X'X\widehat{\beta}) = 0, \\ \frac{\partial \log L}{\partial \sigma^2} \mid (\beta = \widehat{\beta}, \sigma^2 = \widehat{\sigma}^2) = -\frac{T}{2\widehat{\sigma}^2} + \frac{1}{2\widehat{\sigma}^4} (Y - X\widehat{\beta})'(Y - X\widehat{\beta}) = 0. \end{cases}$$

MLE in Multiple Regression

So that we have

$$\begin{aligned}\widehat{\beta} &= (X'X)^{-1}X'Y \\ \widehat{\sigma}^2 &= \frac{1}{T}(Y - X\widehat{\beta})'(Y - X\widehat{\beta})\end{aligned}$$

Now we examine the Hessian matrix

$$H = \frac{\partial^2 \log L}{\partial \theta \partial \theta'} \bigg|_{(\widehat{\beta}, \widehat{\sigma}^2)} = \begin{bmatrix} \frac{\partial^2 \log L}{\partial \beta \partial \beta'} & \frac{\partial^2 \log L}{\partial \beta \partial \sigma^2} \\ \frac{\partial^2 \log L}{\partial \sigma^2 \partial \beta'} & \frac{\partial^2 \log L}{\partial \sigma^2 \partial \sigma^2} \end{bmatrix} \bigg|_{\widehat{\beta}, \widehat{\sigma}^2} = \begin{bmatrix} -\frac{1}{\widehat{\sigma}^2} X'X & 0 \\ 0 & -\frac{T}{2\widehat{\sigma}^4} \end{bmatrix}$$

It is easy to see that it is a negative definite matrix. Thus we have completed the derivation of MLE.

MLE in MA(1) Model

Suppose the model is

$$y_t = \varepsilon_t + \beta \varepsilon_{t-1}$$

and suppose that $\{\varepsilon_t\}$ are white noise with normal distribution (then they are i.i.d.), then the likelihood function for $\varepsilon_1, \dots, \varepsilon_T$ is

$$L = \prod_{t=1}^T \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{\varepsilon_t^2}{2\sigma^2}\right)$$

and taking logarithm

$$\log L = -\frac{T}{2} \log(2\pi) - \frac{T}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{t=1}^T \varepsilon_t^2$$

MLE in MA(1) Model

If we knew the value of β and $\varepsilon_0 = 0$, we could construct $\varepsilon_1, \dots, \varepsilon_T$ recursively by

$$\varepsilon_1 = y_1$$

$$\varepsilon_2 = y_2 - \beta\varepsilon_1 = y_2 - \beta y_1$$

$$\varepsilon_3 = y_3 - \beta\varepsilon_2 = y_3 - \beta(y_2 - \beta y_1)$$

$$\vdots$$

In general, if $|\beta| < 1$ (then the process is **invertible**), we have

$$\varepsilon_t = y_t / (1 + \beta L) = \sum_{i=0}^{t-1} (-\beta)^i y_{t-i}$$

MLE in MA(1) Model

Thus

$$\log L = \frac{-T}{2} \log(2\pi) - \frac{T}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{t=1}^T \left(\sum_{i=0}^{t-1} (-\beta)^i y_{t-i} \right)^2$$

Remark 1: We can use the same procedure as in the multiple regression to derive the MLE estimator. However, in this case, it is not easy to obtain a closed form solution and usually some numerical optimization procedures will be adopted. For general MA(q) or ARMA(p,q) model, it could be more complicated. And when the likelihood function is flat, it is not easy to find the solution.

MLE in MA(1) Model

Remark 2 (invertibility): We have known that if the process is stationary AR process and ARMA process can be written as convergent MA process (possibly $MA(\infty)$). A process $\{y_t\}$ is invertible if can be represented by finite order or convergent autoregressive process. For ARMA(p,q) model (1), the condition for invertibility is that the roots of the polynomials $(1 + \beta_1 L + \cdots + \beta_q L^q)$ must lie outside the unit circle.

Remark 3: It is not difficult to use statistical softwares (e.g., R) to obtain the estimators (the command in R is `arma()`).

Model Selection Criteria

In a model, if we include more lags of y_t or ε_t , we necessarily reduce the sum of squares of residuals. So that there needs to be a trade off between reduction of sum of squares of residuals and a more **parsimonious** model.

Two most commonly used model selection criteria: Akaike Information Criterion (AIC) and the Schwartz Bayesian Criterion (SBC).

- 1 $AIC = T \log(\sum_{t=1}^T \widehat{\varepsilon}_t^2) + 2n$ where n is the number of estimated parameters ($p + q + 1$) and T is the sample size
- 2 $SBC = T \log(\sum_{t=1}^T \widehat{\varepsilon}_t^2) + n \log(T)$

We will choose the model with least AIC or SBC. Notice that in different statistical softwares, they may have different forms.

Testing for White Noise

We can test whether the process $\{y_t\}$ is a white noise by testing whether $\rho_1 = \dots = \rho_s = 0$ (Note that P.68 mentioning test for r_k , $k = 1, \dots, s$ is not correct). We can construct the following statistics (Box-pierce statistics)

$$Q = T \sum_{k=1}^s r_k^2 \rightarrow_d \chi^2(s)$$

The idea is that under the null of white noise, ρ_k $k = 1, \dots, s$ are all zero, so that their estimates will be very small. While when some ρ_k are large, the Q statistic will go to infinity.

A finite sample improvement (Ljung-Box statistic)

$$Q = T(T+2) \sum_{k=1}^s r_k^2 / (T-k) \rightarrow_d \chi^2(s)$$

Box-Jenkins Three-stage Modelling Procedure

Box and Jenkins popularize a three-stage procedure for modelling univariate time series.

- 1 **Identification stage.** Look at the time series plot, the ACF, the PACF
- 2 **Estimation stage.** Each of the tentative models is fitted and the various a_i and β_i coefficients are examined. The goal is to select a stationary and parsimonious model that has a good fit
- 3 **Diagnostic checking.** To ensure that the residuals from the the estimated model mimic a white noise process

Box-Jenkins Three-stage Modelling Procedure

Remark 1: In econometrics analysis, usually there is another stage before these three stages: unit root test (ADF test, PP test) and detrend.

Remark 2: For stage 3, when we test the residuals of an ARMA(p, q) model, Q has an asymptotic χ^2 distribution with degree of freedom $s - p - q$ (if constant is included, $s - p - q - 1$).

Forecast in AR(1) Model

Consider the AR(1) model

$$y_t = a_0 + a_1 y_{t-1} + \varepsilon_t$$

So that we can forecast y_{t+1} conditional on the information up to time t

$$E_t y_{t+1} = a_0 + a_1 y_t$$

And furthermore

$$E_t y_{t+2} = a_0 + a_1 E_t y_{t+1} = a_0 + a_1 (a_0 + a_1 y_t)$$

And we can obtain j -step-ahead forecast by forward iteration

$$E_t y_{t+j} = a_0(1 + a_1 + \cdots + a_1^{j-1}) + a_1^j y_t$$

which is called **forecast function** expressing all the j -step-ahead forecast as a function of the information set in period t .

Forecast in AR(1) Model

Remark: when $|a_1| < 1$, as $j \rightarrow \infty$, $E_t y_{t+j} \rightarrow \frac{a_0}{1-a_1}$. For any stationary ARMA model, the conditional forecast of y_{t+j} converges to unconditional mean as $j \rightarrow \infty$. We can define the j -step-ahead forecast error as the difference between the realized (true) value and the forecasted value

$$e_t(j) \equiv y_{t+j} - E_t y_{t+j}$$

So that we have

$$e_t(1) = y_{t+1} - E_t y_{t+1} = \varepsilon_{t+1}$$

and the j -step-ahead forecast error is

$$e_t(j) = y_{t+j} - E_t y_{t+j} = \varepsilon_{t+j} + a_1 \varepsilon_{t+j-1} + \cdots + a_1^{j-1} \varepsilon_{t+1}$$

Forecast in AR(1) Model

Remark 1: It is easy to see that the mean of the forecast error is zero, which suggest suggests the forecasts are unbiased.

Remark 2: The variance of the forecast error is

$$\text{var}[e_t(j)] = \sigma^2[1 + a_1^2 + \cdots + a_1^{2(j-1)}]$$

So that the variance of the forecast error is an increasing function of j . As such, we have more confidence in short term forecast than the long term forecast. As $j \rightarrow \infty$, the forecast error variance converges to $\frac{\sigma^2}{1-a_1^2}$.

Remark 3: If we further assume that $\{\varepsilon_t\}$ are normally distributed (so that they are i.i.d.), then we can construct 95% confidence interval for one-step-ahead forecast as

$$a_0 + a_1 y_t \pm 1.96\sigma$$

Forecast in Higher Order Models

Consider ARMA(2,1) model

$$y_t = a_0 + a_1 y_{t-1} + a_2 y_{t-2} + \varepsilon_t + \beta_1 \varepsilon_{t-1}$$

So that

$$E_t y_{t+1} = a_0 + a_1 y_t + a_2 y_{t-1} + \beta_1 \varepsilon_t$$

$$E_t y_{t+2} = a_0 + a_1 E_t y_{t+1} + a_2 y_t = a_0(1+a_1) + (a_1^2 + a_2)y_t + a_1 a_2 y_{t-1} + a_1 \beta_1 \varepsilon_t$$

And finally

$$E_t y_{t+j} = a_0 + a_1 E_t y_{t+j-1} + a_2 y_{t+j-2} \text{ for all } j \geq 2$$

$$\begin{aligned} e_t(j) &= a_1(y_{t+j-1} - E_t y_{t+j-1}) + a_2(y_{t+j-2} - E_t y_{t+j-2}) + \varepsilon_{t+j} + \beta_1 \varepsilon_{t+j-1} \\ &= a_1 e_t(j-1) + a_2 e_t(j-2) + \varepsilon_{t+j} + \beta_1 \varepsilon_{t+j-1} \end{aligned}$$

Remark: The forecast will converge to the long-run mean as $j \rightarrow \infty$.

Forecast Evaluation

Typically, there will be several plausible models to forecast and the one with best fit may not forecast best. Consider an ARMA(2,1) model, if we could forecast with the **true** model, then

$$e_T(1) = y_{T+1} - a_0 - a_1 y_T - a_2 y_{T-2} - \beta_1 \varepsilon_T = \varepsilon_{T+1}$$

Remark 1: The the forecast error is the pure *unforecastable* portion of y_{T+1} , no other ARMA model can provide superior forecasting performance.

Remark 2: In practice, we do not know the order of the ARMA process or the true values of the coefficients $\implies$ we estimate coefficients from ARMA model $\implies$ existence of coefficient uncertainty (parameter uncertainty).

Forecast Evaluation

When the parameters are estimated and error terms replaced by residuals

$$E_T y_{t+1} = \widehat{a}_0 + \widehat{a}_1 y_T + \widehat{a}_2 y_{T-1} + \widehat{\beta}_1 \widehat{\varepsilon}_T$$

and

$$e_T(1) = y_{T+1} - (\widehat{a}_0 + \widehat{a}_1 y_T + \widehat{a}_2 y_{T-1} + \widehat{\beta}_1 \widehat{\varepsilon}_T)$$

Remark 1: When we forecast using the estimated model, parameter uncertainty increases as the model becomes more complex (possibly an estimated AR(1) model can be better than estimated ARMA(2,1) model when the true model is ARMA(2,1)) because larger model usually contain more in-sample errors that induce forecast errors.

Forecast Evaluation

Remark 2: How we measure which model has better forecast performance? Hold back a portion of observations from the estimation process and we can estimate the alternative models over the shortened span of data and use these estimates to forecast the observations of the **holdback** period.

Example: Consider an AR(1) model and MA(1) model with 150 observations. Use the first 100 observations to estimate both models and then forecast the 101st value and then derive the forecast error (since we have actual value of 101st observation); and then estimate the AR(1) and MA(1) model using the first 101 observations to estimate the model and then forecast the 102nd observation and derive the forecast error. Let $\{f_{1j}\}$ and $\{f_{2i}\}$ denote the respective forecast. And then do regression of the true value on the forecast value and compare the variance of the residuals.

Forecast Evaluation

Usually, researchers would select the model with the smallest mean squared prediction error (MSPE). Suppose we have H one-step-ahead forecast from 2 different models and denote e_{1i} and e_{2i} as the forecast error series from the two models. Then the MSPE for model 1 is

$$MSPE = \frac{1}{H} \sum_{i=1}^H e_{1i}^2$$

and construct the F statistics

$$F = \frac{\sum_{i=1}^H e_{1i}^2}{\sum_{i=1}^H e_{2i}^2}$$

Forecast Evaluation

Under assumptions

- A1 The forecast errors have zero mean and are normally distributed
- A2 The forecast errors are serially uncorrelated
- A3 The forecast errors are contemporaneously uncorrelated with each other

then the F statistics has a standard F distribution with (H, H) degree of freedoms.

Remark: The assumptions are too restrictive in practice.

The Granger-Newbold Test

To overcome the problem of contemporaneously correlated forecast errors (A3), Granger and Newbold (1976) proposed a test statistics as follows

- 1 construct two sequence based on the forecast errors:

$$x_i = e_{1i} + e_{2i} \text{ and } z_i = e_{1i} - e_{2i}, i = 1, 2, \dots, H$$

- 2 Then we have $\rho_{xz} = Ex_i z_i = (e_{1i}^2 - e_{2i}^2)$

- 3 Let r_{xz} denote the sample correlation coefficient if Assumptions 1 and 2 hold, then

$$\frac{r_{xz}}{(1 - r_{xz}^2)/(H - 1)}$$

has a t -distribution with $H - 1$ degree of freedom.

The Diebold-Mariano Test

To overcome the problem of (A1) and (A2), Diebold and Mariano (1995) developed a test relaxing assumptions (A1), (A2) and (A3).

As before, we are concerned about one-step-ahead forecast. And we use different loss function rather than quadratic form and define $d_i = g(e_{1i}) - g(e_{2i})$. the mean loss function can be obtained as

$$\bar{d} = \frac{1}{H} \sum_{i=1}^H [g(e_{1i}) - g(e_{2i})]$$

Under some regularity condition, it can be shown that $\frac{\bar{d}}{\sqrt{\text{var}(\bar{d})}}$ converges asymptotically to $N(0,1)$.

The Diebold-Mariano Test

- Case 1** When the $\{d_i\}$ are serial uncorrelated, the estimate of $\text{var}(\bar{d})$ is simply $\gamma_0/(H-1)$
- Case 2** when the first q values of γ_i are different from zero, then $\text{var}(\bar{d})$ can be approximated by $(\gamma_0 + 2\gamma_1 + \cdots + 2\gamma_q)/(H-1)$
- Case 3** When q is unknown, the robust **long run variance** estimator will be obtained using standard method, e.g. Newey and West (1987).

Remark: Procedures for multi-step-ahead forecasts can be obtained similarly.

Combining Forecasts

We have discussed how to choose the ‘best’ model to forecast, however, it is possible to forecast with all the plausible model and then take the average of the forecasts.

Let the series $\{f_{yt}\}$ denote the one-step-ahead forecast of y_t from model i , $i = 1, \dots, n$. Then we can make the composite forecast f_{ct} constructed as follows

$$f_{ct} = w_1 f_{1t} + \dots + w_n f_{nt}$$

where $\sum_{i=1}^n w_i = 1$.

Combining Forecasts

Remark 1: It is easy to see that if the forecasts are unbiased, then the composite forecast is also unbiased (note that the unbiased definition in p.109 is wrong, it should be $E_{t-1}(y_t - f_{it}) = 0$).

Remark 2: if $n = 2$, the variance of the composite forecast error is $var(e_{ct}) = w_1^2 var(e_{1t}) + w_2^2 var(e_{2t}) + 2w_1(1 - w_1)cov(e_{1t}, e_{2t})$.

Suppose they are uncorrelated with same variance, letting $w_1 = 0.5$, then we have $var(e_{ct}) = 0.5var(e_{1t}) = 0.5var(e_{2t})$ (note there is typo in p.109, 0.5 not 0.25).

Remark 3: An optimal weight can be chosen to minimize the variance (similar to the portfolio choice problem) of forecast error with

$$w_1^* = \frac{var(e_{2t}) - cov(e_{1t}, e_{2t})}{var(e_{1t}) + var(e_{2t}) - 2cov(e_{1t}, e_{2t})}$$

课外阅读

Hansen, B.E., 2007. Least Squares Model Averaging. *Econometrica*, 75, 1175-1189.

Aruoba, S.B. and Diebold, F.X., 2010. Real-Time Macroeconomic Monitoring: Real Activity, Inflation, and Interactions. *American Economic Review*, 100, 20-24.

The Interest Rate Spread

The textbook discusses the ‘textbook example’ to illustrate the application of the methodology.

The data is quarterly data of the spread ($\{s_t\}$) between a long term (interest rate on 5-year U.S. government bonds) and short term interest rates (3-month treasury bill) from 1960Q1 to 2012Q4.

Time Series Plot

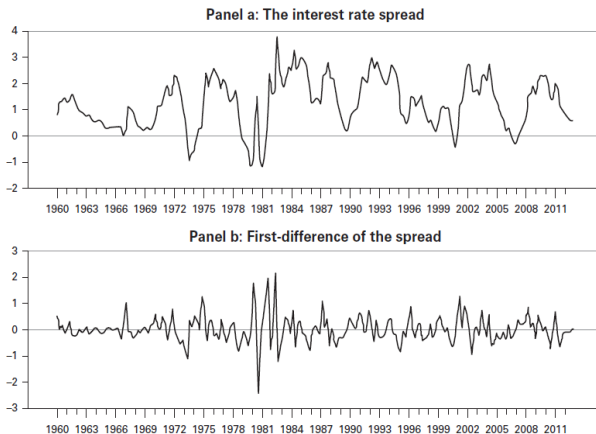


FIGURE 2.5 Time Path of the Interest Rate Spread

ACF and PACF

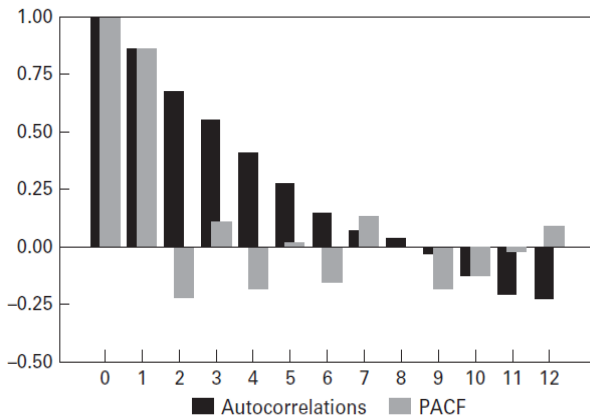


FIGURE 2.6 ACF and PACF of the Spread

Estimation and Model Selection

Table 2.4 Estimates of the Interest Rate Spread

	AR (7)	AR (6)	AR (2)	$p = 1, 2, 7$	ARMA (1,1)	ARMA (2,1)	$p = 2;$ $ma = (1, 7)$
a_0	1.20 (6.57)	1.20 (7.55)	1.19 (6.02)	1.19 (6.80)	1.19 (6.16)	1.19 (5.56)	1.20 (5.74)
a_1	1.11 (15.76)	1.09 (15.54)	1.05 (15.25)	1.04 (14.83)	0.76 (14.69)	0.43 (2.78)	0.36 (3.15)
a_2	-0.45 (-4.33)	-0.43 (-4.11)	-0.22 (-3.18)	-0.20 (-2.80)		0.31 (2.19)	0.38 (3.52)
a_3	0.40 (3.68)	0.36 (3.39)					
a_4	-0.30 (-2.70)	-0.25 (-2.30)					
a_5	0.22 (2.02)	0.16 (1.53)					
a_6	-0.30 (-2.86)	-0.15 (-2.11)					
a_7	0.14 (1.93)			-0.03 (-0.77)			
β_1					0.38 (5.23)	0.69 (5.65)	0.77 (9.62)
β_7							-0.14 (-3.27)
SSR	43.86	44.68	48.02	47.87	46.93	45.76	43.72
AIC	791.10	792.92	799.67	801.06	794.96	791.81	784.46
SBC	817.68	816.18	809.63	814.35	804.93	805.10	801.07
$Q(4)$	0.18	0.29	8.99	8.56	6.63	1.18	0.76
$Q(8)$	5.69	10.93	21.74	22.39	18.48	12.27	2.60
$Q(12)$	13.67	16.75	29.37	29.16	24.38	19.14	11.13

Notes:

To ensure comparability, each equation was estimated over the 1961Q4 – 2012Q4 period. Values in parentheses are the t -statistics for the null hypothesis that the estimated coefficient is equal to zero. SSR is the sum of squared residuals. $Q(n)$ are the Ljung–Box Q -statistics of the residual autocorrelations.

For ARMA models, many software packages do not actually report the intercept term a_0 . Instead, they report the estimated mean of process, μ_y , along with the t -statistic for the null hypothesis that $\mu_y = 0$. The historical reason for this convention is that it was easier to first demean the data and then estimate the ARMA coefficients than to estimate all values in one step. If your software package reports a constant term approximately equal to 0.216, it is reporting the estimated intercept.

Estimation, Model Selection and White Noise Test

From the results, the AIC selects AR(7) while the SBC selects AR(2) among the AR models. However, Q-statistics for the residual indicates it is not white noise (p.92).

From the sample ACF and PACF, it seem that ARMA type model should be better that AR type model. And the AIC selects ARMA(2,1) over the AR(6) while the SBC selects ARMA(2,1) over AR(6) or AR(7). The values for $Q(4)$, $Q(8)$ and $Q(12)$ indicate that the autocorrelations in the residuals are not statistically significant at the 5% level.

From the ACF of the residual from ARMA(2,1) (p.92), we can see that the autocorrelation at lag 7 is large. So that ARMA(2,7) is tried. ARMA(2,[1,7]) are good models by information criteria and Q test.

Out-of-Sample Evaluation

Total number of observations is 205, hold back period contains 50 observations.

According to Ganger-Newbold test, $AR(7)$ and $ARMA(2,[1,7])$ are not significantly different.

Using DM test, there is evidence in favor of $ARMA(2,[1,7])$ model.

Using seven models (p.111), we can do composite forecast and find better forecasting models.

Seasonality

Many economic processes exhibit some form of seasonality, E.g.: the agricultural sector depends on the weather; the retail trade in Thanksgiving-to-Christmas holiday season...

Many widely used time series data are **deseasonalized or seasonally adjusted**

Two purely seasonal models for quarterly data

$$y_t = a_4 y_{t-4} + \varepsilon_t \mid |a_4| < 1$$

and

$$y_t = \beta_4 \varepsilon_{t-4} + \varepsilon_t$$

Seasonality

Remark: For the seasonal AR(1) model, $\rho_i = (a_4)^{i/4}$ if $i/4$ is an integer and $\rho_i = 0$, otherwise. For the seasonal MA(1) model, the ACF exhibits a single peak at lag 4, and all other correlations are 0.

Multiplicative seasonality allows for interaction of the ARMA and the seasonal effects. E.g.,

$$(1 - a_1 L)y_t = (1 + \beta_1 L)(1 + \beta_4 L^4)\varepsilon_t$$

This equation can be rewritten as

$$y_t = a_1 y_{t-1} + \varepsilon_t + \beta_1 \varepsilon_{t-1} + \beta_4 \varepsilon_{t-4} + \beta_1 \beta_4 \varepsilon_{t-5}$$

Seasonal Difference

Some time series data, e.g., the logarithm of M1, denoted by $\{y_t\}$, have strong seasonal persistence pattern. So that usually seasonal difference is taken ($y_t - y_{t-4}$ for quarterly data). We can transform the data

$$m_t = (1 - L)(1 - L^4)y_t$$

Then we can use ARMA model with seasonality to model $\{m_t\}$ sequence. You can read the textbook for the analysis of the money supply data.

Testing for Structure Change

Application of Box-Jenkins methodology is that the structure of DGP does not change. The values of a_i and β_i should be constant from one period to the next.

The **Chow Test** is the natural tool for testing structure break. Split T observations into two subsamples with t_m observation in the first subsample and $T - t_m$ in the second subsample. And the estimate ARMA(p,q) model using the two subsamples. Denote the sum of the squared residuals from each model by SSR_1 and SSR_2 respectively. To test for the null $a_i(1) = a_i(2)$, $i = 0, \dots, p$ and $\beta_j(1) = \beta_j(2)$, $j = 1, \dots, q$, use the F -test

$$F = \frac{(SSR - SSR_1 - SSR_2)/n}{(SSR_1 + SSR_2)/(T - 2n)}$$

where n is the number of parameters estimated.

Endogenous Breaks and Parameter Instability

Remark 1: In the Chow test, the breaking day is prespecified, which is not always suitable. E.g., it is not clear how we can provide a specific break date to the development of advent of 'financial deregulation'.

Remark 2: When it is not prespecified, there is **endogenous break**. In this case, the tests like Andrews and Ploberger (1994) and Hansen (1997) will be applied. Another popular test for parameter stability is the CUSUM test of Brown, Durbin and Evans (1975).

Summary and Conclusions

This chapter focuses on the Box-Jenkins approach to identification (time series plot, ACF, PACF), estimation (OLS or MLE), diagnostic checking (model selection and testing) and forecasting a univariate time series.

Two additional topics-seasonality and structure break are discussed.

The Box-Jenkins methodology is an art rather than science (also true for most empirical analysis) and it is up to the researchers to determine the 'best' models.

Homework

Exercise 2, 5

The homework must be submitted one week later.

Time Series Analysis

Lecture 4: Modeling Volatility

Biqing Cai
Associate Professor

School of Economics, HUST

Spring, 2024

Outline of This Lecture

- 1 Introduction
- 2 ARCH and GARCH Modelling
- 3 Some Extensions
- 4 Estimating the NYSE U.S. 100 Index
- 5 Summary and Conclusions
- 6 Homework

Why We Need to Model Volatility?

In Lecture 3, we have studied the ARMA model which is concerned about the conditional mean (first moment) of time series.

Volatility (conditional variance, conditional heteroskedasticity) modelling is also of crucial importance in some economic or financial problems.

Example: In the classic Markowitz (1952) portfolio theory, people choose the portfolio with the minimized variance among the portfolios with the same mean (or the portfolio with maximized mean with the same variance).

Furthermore, since Engle (1982), it is well known that the so-called *volatility clustering* is a stylized fact for economic time series.

Stylized Facts in Economic Time Series

In the development of econometric time series, several stylized facts have been summarized.

1. **Many of the series contain a clear trend.** Real and potential U.S. GDP, consumption and investment exhibit a decidedly upward trend (Figure 3.1).
2. **The volatility of many time series is not constant over time.** The volatility of real GDP appears to fall in 1984, shows a negative spike in 2007, then stabilize (Figure 3.2).
3. **Shocks to a series can display a high degree of persistence.** Short term and long term interest rates (Figure 3.4).

Stylized Facts in Economic Time Series

4. **Some of the series seem to meander.** The exchange rates go through sustained periods of appreciation and depreciation without tendency to revert to long-run mean: random walk behavior (Figure 3.5).
5. **Some series share comovements with other series.** Short term and long term interest rates (Figure 3.4): common stochastic trend, cointegration.
6. **Some of the series exhibit breaks.** Oil price after the financial crisis (Figure 3.6).

This chapter is concerned about the 2nd stylized fact: volatility clustering. And the most widely used tool to analyze the phenomenon is the ARCH/GARCH type models.

The GDP Growth Rate

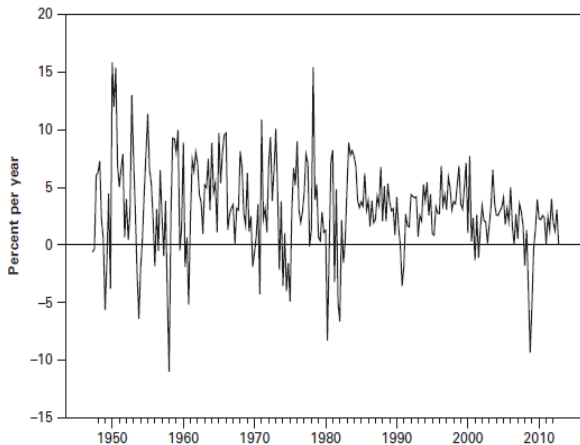


FIGURE 3.2 Annualized Growth Rate of Real GDP

The NYSE Stock Return

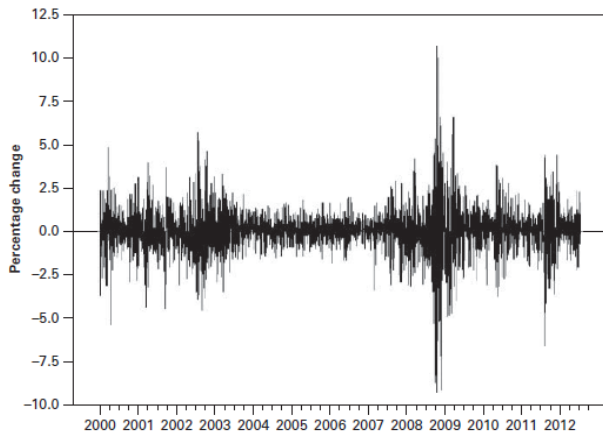


FIGURE 3.3 Percentage Change in the NYSE U.S. 100 (January 4, 2000–July 16, 2012)

Preliminary Analysis

Remark 1: As having been emphasized in the stylized fact 2, the assumption of constant variance (homoskedasticity) seem to be inappropriate. One approach to catch the effect is $y_{t+1} = \varepsilon_{t+1}x_t$ where y_{t+1} is the variable of interest (demeaned variable), ε_{t+1} is white noise with variance σ^2 and x_t is a state variable of period t . So that $\text{var}(y_{t+1} | F_t) = \sigma^2 x_t^2$

Remark 2: From the two figures, we can also see that the volatility is clustering, i.e., high volatility is accompanied by high volatility and vice versa. So that it seems appropriate to select x_t^2 to be the volatility of period t .

Preliminary Analysis

Remark 3: From Figure 3.2, we can see that the volatility decrease after 1980s. This is called “Great Moderation” in economics. The Great Moderation is the name given to the period of decreased macroeconomic volatility experienced in the U.S. since the 1980s. I refer to Stock and Watson (2002, NBER Macroeconomics Annual) for an excellent survey.

Remark 4: From Figure 3.3, we can see that the volatility was high between 2008-2010 because of the financial crisis.

ARCH Models

We considered an ARCH(1) model which is a special case of the general ARCH models proposed in Engle (1982).

$$\varepsilon_t = v_t \sqrt{\alpha_0 + \alpha_1 \varepsilon_{t-1}^2},$$

where $\{v_t\}$ is i.i.d. process with mean 0 and variance 1, α_0 and α_1 are constants such that $\alpha_0 > 0$ and $0 \leq \alpha_1 < 1$.

Remark 1: $\{\varepsilon_t\}$ is a m.d.s., so that its mean is 0 and it is serial uncorrelated. And because $0 \leq \alpha_1 < 1$, the process is stationary, so that $E\varepsilon_t^2 = \alpha_0/(1 - \alpha_1)$.

ARCH Models

Remark 2: And we have $E(\varepsilon_t^2 | F_{t-1}) = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2$, i.e., the conditional variance of ε_t is dependent on the realized value of ε_{t-1}^2 . Or more specifically, the conditional variance is a first order autoregressive process. If ε_{t-1}^2 is large, the conditional variance of ε_t will be large as well (volatility clustering).

Remark 3: We can combine the conditional mean model (ARMA model) and the conditional variance model to form more general model.

$$y_t = a_0 + \sum_{i=1}^p a_i y_{t-i} + \sum_{i=0}^q \beta_i \varepsilon_{t-i}$$
$$\varepsilon_t = v_t \sqrt{\alpha_0 + \sum_{i=1}^m \alpha_i \varepsilon_{t-i}^2}$$

GARCH Model

Similar to the extension of AR model to ARMA model, Bollerslev (1986) generalized the ARCH model to GARCH model. A GARCH(p,q) model is as follows

$$\begin{aligned}\varepsilon_t &= \nu_t \sqrt{h_t} \\ h_t &= \alpha_0 + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^p \beta_j h_{t-j}\end{aligned}$$

where $\{\nu_t\}$ is an i.i.d. process with mean 0 and variance 1. $\alpha_0 > 0$, $\alpha_i \geq 0$, for all $i = 1, \dots, q$, $\beta_j \geq 0$, for all $j = 1, \dots, p$ and $\sum_{i=1}^q \alpha_i + \sum_{j=1}^p \beta_j < 1$.

Remark 1: When $p = 0$, the process becomes an ARCH(q) process. Similar to ARMA model in comparison to AR model, a GARCH type model may be more parsimonious than ARCH model.

GARCH Model

Remark 2: $\{\varepsilon_t\}$ is a m.d.s., so that its mean is 0 and it is serial uncorrelated.

Remark 3: $E(\varepsilon_t^2 | F_{t-1}) = \alpha_0 + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^p \beta_j h_{t-j}$. And because $\sum_{i=1}^q \alpha_i + \sum_{j=1}^p \beta_j < 1$, the process is stationary, so that $E\varepsilon_t^2 = \alpha_0 / (1 - \sum_{i=1}^q \alpha_i - \sum_{j=1}^p \beta_j)$.

Detecting ARCH/GARCH Effects

The key feature of ARCH/GARCH models is that conditional variance of the disturbance (innovation) of the $\{y_t\}$ sequence acts like an AR/ARMA process. Hence we expect that the residuals from a fitted ARMA model (conditional mean model) should display thus characteristic pattern.

More specifically, after fitting ARMA model for $\{y_t\}$ sequence, if the residuals are white noise process (Box-Ljung test, ACF, PACF), then the conditional mean model is adequate. If there is further conditional heteroskedasticity, we expect to find serial correlation in the *squared* residuals.

Detecting ARCH/GARCH Effects

The procedure for detecting conditional heteroskedasticity (McLeod and Li 1983).

- 1 Estimate a ‘best fitting’ ARMA model for the $\{y_t\}$ sequence and obtain the residual sequence $\{\widehat{\varepsilon}_t\}$ sequence. Calculate the sample variance by $\widehat{\sigma}^2 = \frac{\sum_{t=1}^T \widehat{\varepsilon}_t^2}{T}$ where T is the number of residuals.
- 2 Calculate and plot the sample autocorrelations of the squared residuals with $r_i = \frac{\sum_{t=i+1}^T (\widehat{\varepsilon}_t^2 - \widehat{\sigma}^2)(\widehat{\varepsilon}_{t-i}^2 - \widehat{\sigma}^2)}{\sum_{t=1}^T (\widehat{\varepsilon}_t^2 - \widehat{\sigma}^2)}$.
- 3 If there is no conditional heteroskedasticity, then under some regularity conditions, $\sqrt{T}r_i$ will converge in distribution to a standard normal variable. Or we can use the Box-Ljung test $Q = T(T+2) \sum_{i=1}^m r_i^2 / (T-i)$ which has an asymptotic χ^2 distribution with m degrees of freedom under the null.

Detecting ARCH/GARCH Effects

A more formal procedure Lagrange multiplier test for ARCH effect by Engle (1982).

- 1 Estimate a ‘best fitting’ ARMA model for the $\{y_t\}$ sequence and obtain the residual sequence $\{\widehat{\varepsilon}_t\}$ sequence.
- 2 Regress the squared residuals on a constant and q lags of the values $\widehat{\varepsilon}_{t-1}^2, \dots, \widehat{\varepsilon}_{t-q}^2$, i.e.,

$$\widehat{\varepsilon}_t^2 = \alpha_0 + \sum_{j=1}^q \alpha_j \widehat{\varepsilon}_{t-j}^2 + \eta_t$$

If there are no ARCH/GARCH effects (under the null), the estimated values for α_i , $i = 1, \dots, q$ should be close to zero such that the coefficient of determinant (R^2) should be close to 0. The test statistics $L = TR^2$ has an asymptotic χ^2 distribution with q degrees of freedom under the null.

MLE Estimation of GARCH Models

Consider the process is

$$y_t = \beta' x_t + \varepsilon_t$$

$$\varepsilon_t = \nu_t \sqrt{h_t}$$

$$h_t = \alpha_0 + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^p \beta_j h_{t-j}$$

where $\{x_t\}$ is a vector of regressors. We further assume that $\{\nu_t\}$ is a sequence of i.i.d. standard Normal process.

Remark: In the GARCH model literature, ν_t is usually assumed to be either normally distributed or t -distributed (to catch the heavy tail). When the true distribution is not normal, but we take the normality assumption, it is called the **quasi-MLE (QMLE)** estimation.

MLE Estimation of GARCH Models

As we have discussed in Lecture 3, the likelihood function is

$$L = \prod_{t=1}^T \left(\frac{1}{2\pi h_t} \right) \exp \left(\frac{-\varepsilon_t^2}{2h_t} \right)$$

Taking logarithm

$$\log L = -\frac{T}{2} \log(2\pi) - 1/2 \sum_{t=1}^T \log h_t - 1/2 \sum_{t=1}^T (\varepsilon_t^2 / h_t)$$

and substitute ε_t with $y_t - \beta' x_t$, h_t with

$$\alpha_0 + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^p \beta_j h_{t-j}.$$

Remark: It is difficult to obtain close form solution and usually numerical methods are required.

Assess the Fitting

Similar to ARMA modelling procedure, we can assess the fit of a GARCH model using model selection criteria such as AIC and SBC, where

$$AIC = -2 \log L + 2n$$

$$SBC = -2 \log L + n(\log T)$$

where n is the number of estimated parameter. The model with smaller AIC or SBC should be regarded as better model.

Diagnostic Checks for Model Adequacy

Estimate the conditional mean model (ARMA) model and get the residual sequence $\{\widehat{\varepsilon}_t\}$, then estimate the conditional variance model and get the estimated $\{\widehat{h}_t\}$ sequence and finally construct the sequence $s_t = \frac{\widehat{\varepsilon}_t}{\widehat{h}_t^{1/2}}$ (which is the standard residual). The test of model adequacy falls into two parts.

- 1 Testing whether the sequence $\{s_t\}$ is white noise process (using Ljung-Box Q statistics)
- 2 Testing whether the sequence $\{s_t^2\}$ contain conditional heteroskedasticity (using Engle's LM test)

Forecasting

When we have a satisfactory model for $\{y_t\}$ sequence, we can forecast future values of y_t and its conditional variance.

The one-step-ahead forecast is $E_t y_{t+1}$, because the error term is still m.d.s. in presence of conditional heteroskedasticity, the forecast is the same as the forecast of pure ARMA model.

However, when there is conditional heteroskedasticity, $var_t(y_{t+1}) = h_{t+1}$, so that the confidence interval for the forecast is now given by

$$E_t y_{t+1} \pm 1.96 h_{t+1}^{1/2}$$

Forecasting

Consider a GARCH(1,1) model, if we forecast the one-step-ahead variance, we have

$$h_{t+1} = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 h_t$$

For the j -step-ahead forecast, we have $(\{\nu_t\})$ i.i.d. with mean 0 and variance 1)

$$E_t \varepsilon_{t+j}^2 = E_t h_{t+j}$$

and

$$E_t h_{t+j}^2 = E_t (\alpha_0 + \alpha_1 \varepsilon_{t+j-1}^2 + \beta_1 h_{t+j-1}) = \alpha_0 + (\alpha_1 + \beta_1) E_t h_{t+j-1}$$

Continuing the derivation leads to

$$E_t h_{t+j} = \alpha_0 [1 + (\alpha_1 + \beta_1) + \cdots + (\alpha_1 + \beta_1)^{j-1}] + (\alpha_1 + \beta_1)^j h_t$$

Forecasting

Remark 1: If $(\alpha_1 + \beta_1) < 1$, as $j \rightarrow \infty$ $E_t h_{t+j} \rightarrow \frac{\alpha_0}{1 - \alpha_1 - \beta_1}$ which is the unconditional variance Eh_t .

Remark 2: The procedure also applies for general GARCH(p,q) model.

The IGARCH Model

In financial time series, typically the volatility is quite persistent such that the estimated coefficients of a GARCH(1,1) model usually satisfy $\widehat{\alpha}_1 + \widehat{\beta}_1$ close to 1. So that an integrated-GARCH (IGARCH) (GARCH(1,1) with $\alpha_1 + \beta_1 = 1$) model is usually used to model the phenomenon.

For the IGARCH, the j -step ahead Forecast is $E_t h_{t+j} = j\alpha_0 + h_t$. And the unconditional variance is clearly infinite.

Using the lag operator, the process can be written as $h_t = \alpha_0 + (1 - \beta_1)\varepsilon_{t-1}^2 + \beta_1 Lh_t$ and solving for h_t (when $\beta_1 < 1$)

$$h_t = \alpha_0 / (1 - \beta_1) + (1 - \beta_1) \sum_{i=0}^{\infty} \beta_1^i \varepsilon_{t-1-i}^2$$

h_t is a geometrically decaying function of past $\{\varepsilon_t^2\}$.

The ARCH-M Model

The ARCH in mean (ARCH-M) model allows the mean to depend on its own conditional variance. It is used to model the risk premium.

The model is

$$y_t = \mu_t + \varepsilon_t$$

$$\mu_t = \beta + \delta h_t, \delta > 0$$

$$h_t = \alpha_0 + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2$$

Models with Explanatory Variables

The specification of h_t can allow for exogenous variables.

Example: We want to see whether the terrorist attack of Sep 11, 2001 increased the volatility of the stock market. We consider the following specification

$$h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 h_{t-1} + \gamma D_t$$

where D_t is a dummy variable such that it is equal to 0 before Sep 11, 2001 and it is equal to 1 after Sep 11, 2001. When the estimator of $\gamma > 0$ (and significant in the statistical sense), it is possible to conclude that the terrorist attack increased the volatility.

Models with Asymmetry: TARCh and EGARCH

The **leverage effect** is a stylized fact in financial market.

An interesting feature of asset prices is that bad news seems to have a more pronounced effect on the volatility than does good news. For many stocks, there is a strong negative correlation between current return and the future volatility because of the leverage effect.

The reason is that, a negative shock decreases the equity and increases the leverage (debt-to-equity ratio), thus increases the risk of holding the equity.

Models with Asymmetry: TARCh and EGARCH

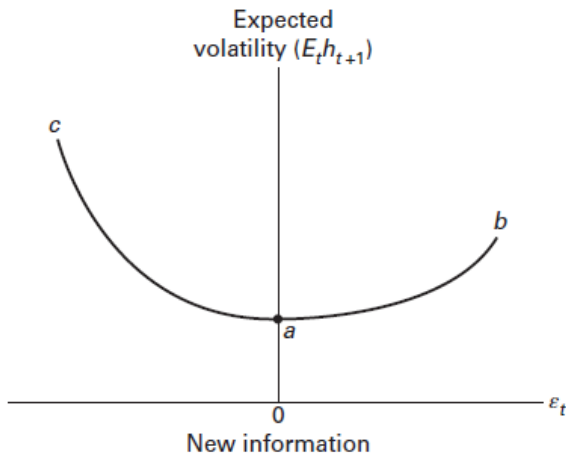


FIGURE 3.11 The Leverage Effect

Models with Asymmetry: TARARCH and EGARCH

The threshold-GARCH (TGARCH) model is used to catch the leverage effect.

$$h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \lambda_1 d_{t-1} \varepsilon_{t-1}^2 + \beta_1 h_{t-1}$$

where d_{t-1} is a dummy variable that is equal to 1 if $\varepsilon_{t-1} < 0$ and is equal to 0 if $\varepsilon_{t-1} \geq 0$.

Remark: When $\lambda_1 > 0$, the model implies that negative shocks have larger effects on the volatility than the positive shocks.

Models with Asymmetry: TARCh and EGARCH

The exponential-GARCH (EGARCH) model can also be used to catch the leverage effect.

$$\log h_t = \alpha_0 + \alpha_1(\varepsilon_{t-1}/h_{t-1}^{1/2}) + \lambda_1 |\varepsilon_{t-1}/h_{t-1}^{1/2}| + \beta_1 \log h_{t-1}$$

Remark: Besides catching the leverage effect, the model allows the coefficients to be negative.

Models with Asymmetry: TARARCH and EGARCH

We now discuss how to test the leverage effect. In the TGARCH model, it corresponds to test whether $\lambda_1 > 0$ and in the EGARCH model, it corresponds to test whether $\alpha_1 = 0$ (typo in p. 157).

Another method of test: construct $\{s_t\}$ sequence via $s_t = \widehat{\varepsilon}_t / \widehat{h}_t^{1/2}$ where $\widehat{\varepsilon}_t$ is the residual from conditional mean model and $\widehat{h}_t$ is the conditional variance from conditional variance model (ARCH or GARCH without leverage). Then test whether $a_1 = \dots = a_p$ in the following regression

$$s_t^2 = a_0 + a_1 s_{t-1} + \dots + a_p s_{t-p} + \eta_t$$

Multivariate GARCH

When we have data set with several variables, we can estimate the conditional volatilities simultaneously (consider the portfolio choice example).

In a multivariate GARCH model, the contemporaneous shocks can be correlated with each other.

Moreover, multivariate GARCH models allows volatility spillovers in that volatility shocks to one variable might affect the volatility of other related variables. E.g., a shock to short term interest rate might also affect long term interest rate.

Multivariate GARCH

Consider the bivariate case.

$$\varepsilon_{1t} = \nu_{1t}(h_{1t})^{1/2}$$

$$\varepsilon_{2t} = \nu_{2t}(h_{2t})^{1/2}$$

where $\nu_t \equiv (\nu_{1t}, \nu_{2t})' \sim i.i.d.(0, \Sigma)$ with $\Sigma = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$. And the

volatility follows the so-called vech model, see equations (3.42)-(3.44) in the textbook (p.166).

Multivariate GARCH

There are drawbacks for the vech model.

- Difficult to estimate and the number of parameters is large
- the conditional variance matrix is not guaranteed to be positive definite in the model.

To circumvent the first problem, one can use the diagonal vech; to circumvent the first problem, one can use the BEKK (1991) model such that

$$H_t = C'C + A'\varepsilon_{t-1}\varepsilon'_{t-1}A + B'H_{t-1}B$$

The drawback of the model: difficult to estimate, large number of parameters.

Multivariate GARCH

Another popular multivariate GARCH model is the constant conditional correlation (CCC) model where

$$h_{12t} = \rho_{12}(h_{11t}h_{22t})^{1/2},$$

which largely decrease the number of parameters estimated and guarantee positive definite of the conditional variance matrix.

Remark: When the correlation is constant, the CCC model is a good choice. However, in applications, the constant correlation assumption is a little bit restricted, so that a dynamic conditional correlation (DCC) model is required.

Multivariate GARCH

We now discuss how to construct the volatility impulse response function. Consider the bivariate volatility model is given by equation (3.42)-(3.44), then the one-step-ahead forecast for h_{11t+1} is

$$\begin{aligned} E_t h_{11t+1} &= c_{10} + \alpha_{11} \varepsilon_{1t}^2 + \alpha_{12} \varepsilon_{1t} \varepsilon_{2t} + \alpha_{13} \varepsilon_{2t}^2 \\ &+ \beta_{11} h_{11t} + \beta_{12} h_{12t} + \beta_{13} h_{22t} \end{aligned}$$

And j -step-ahead forecast can be constructed recursively.

We can also construct the impulse volatility function viz $E_{T^*}(h_{11T+i}) - E_T(h_{11T+i})$ where $E_{T^*}(h_{11T+i})$ denotes we disturb a unit shock to ε_{jT} and then construct forecast.

Data Description

The data is log difference (log return) of NYSE Index of 100 U.S. stocks.

The data is daily and is from Jan 4, 2000 to Jul 16, 2012 with a total of 3270 observations.

The data covers the period 2004-2006 when the volatility is small and 2008-2009 when the volatility is large.

Figure 3.3 gives the time series plot.

Conditional Mean and Variance Modelling

- Applying the Box-Jenkins modelling strategy, we can see that AIC model selects an AR(2) model. And the model is adequate.
- Construct the squared residual sequence and applying the LM test, it is concluded that there exists ARCH/GARCH effect.
- An AR(2)-GARCH(1,1) model is fitted.
- Construct the sequence $s_t = \widehat{\varepsilon}_t/h_t^{1/2}$ and do diagnostic checking. We conclude that there is no further serial correlation and no further GARCH effect.
- The test of leverage effect suggests that there is a leverage effect; A TGARCH model is not suitable with one of the estimated coefficient being negative; An EGARCH model is not adequate by diagnostic checking.

课后阅读

Anderson, T.G., T., Bollerslev, and F.G., Diebold, 2010.
Parametric and Nonparametric Volatility Measurement. In Vol 1,
Handbook of Financial Econometrics, edited by Y., Aït, Sahalia and
L.P., Hansen. Elsevier.

Summary and Conclusions

- Conditional heteroskedasticity is a stylized fact in financial market
- ARCH/GARCH type models (and their extensions) are the main tool to analyze the conditional heteroskedasticity
- Procedure for the model fitting is studied with an illustration using NYSE stock index

Homework

Exercise 3, 7

The homework must be submitted one week later.

Time Series Analysis

Lecture 5: Multi-equation Time Series Models

Biqing Cai

Associate Professor

School of Economics, HUST

Spring, 2024

Outline of This Lecture

- 1 Introduction
- 2 Intervention Analysis
- 3 ADLs and Transfer Functions
- 4 VAR Analysis
- 5 Summary and Conclusions
- 6 Homework

We have studied the conditional mean and variance models of a univariate time series. However, multivariate models are required when the relationships of several variables are of interest.

Example 1: In study of term structure of interest rates, we need to jointly model the dynamics of interest rates with different maturities.

Example 2: A model of income dynamics with money supply as a regressor can help us to see whether money supply has effect on real economy (Keynes v.s. Friedman).

Example 3: According to the Phillips curve, there is a trade-off between high inflation rate and high unemployment rate. We can build a model for the joint dynamics of the two variables.

Intervention Analysis

The U.S. began to install metal detector in airports from Jan 1973 in response to the rise in skyjackings. We want to know whether the policy is efficient.

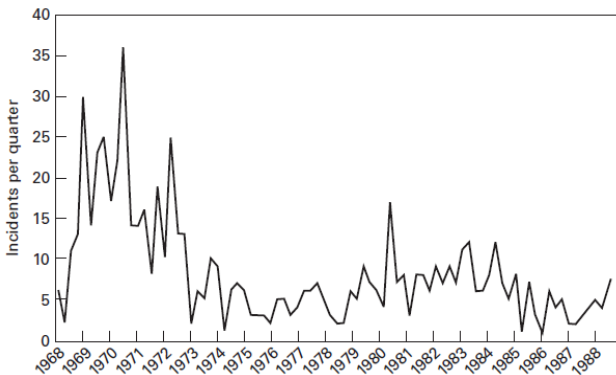


FIGURE 5.2 Skyjackings

Intervention Analysis

From the figure, it seems that the policy is efficient. An intervention analysis allows a formal test.

$$y_t = a_0 + a_1 y_{t-1} + c_0 z_t + \varepsilon_t, \quad |a_1| < 1$$

where y_t denotes the number of skyjackings and z_t is the intervention (dummy) variable that takes value 0 prior to 1973Q1 and 1 beginning in 1973Q1.

Remark 1: Solving the equation, we can obtain

$$y_t = a_0/(1 - a_1) + c_0 \sum_{i=0}^{\infty} a_1^i z_{t-i} \sum_{i=0}^{\infty} a_1^i \varepsilon_{t-i} \quad (1)$$

Intervention Analysis

Remark 2 [Impulse Response Analysis]: Based on (1), we have
(note the difference between the textbook and this lecture)

$$\frac{\partial y_{t+j}}{\partial z_t} = c_0 a_1^j,$$

with $j = 0, 1, \dots$. Also, notice that the effect of a shock on z_t (from 0 to 1) is persistent in the sense that $z_{t+j} = 1$ for all $j \geq 0$. Thus the impulse response analysis is different from what was used in Chapter 1. Denote $l_t(j)$ as the impulse response function, we have

$$l_t(j) = \sum_{i=0}^j \frac{\partial y_{t+i}}{\partial z_{t+i}} = c_0 [1 + a_1 + \dots + a_1^j] \quad (2)$$

Intervention Analysis

Remark 3: From (2), we can see that as $j \rightarrow \infty$, the long-run effect is $c_0/(1 - a_1)$. In Chapter 1, we know that if there is no intervention effect, the long-run mean of the series is $a_0/(1 - a_1)$. After the time of intervention, the drift term now becomes $a_0 + c_0$ and the long-run mean now becomes $(a_0 + c_0)/(1 - a_1)$. So that the long-run effect of intervention is simply the difference of these two terms.

Remark 4: The model can be generalized to ARMA(p,q) intervention model as

$$y_t = a_0 + A(L)y_{t-1} + c_0 z_t + B(L)\varepsilon_t$$

Different Types of Intervention Function

The follow figure gives four different types of intervention functions.

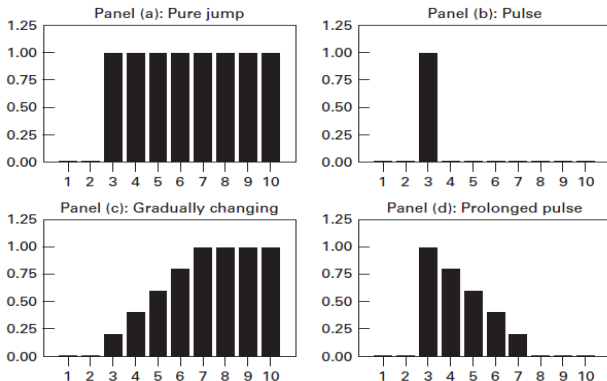


FIGURE 5.3 Typical Intervention Functions

Estimating the Effect of Metal Detector

Before the analysis of intervention effect, a pretest of the data is a good choice, i.e., we fit the ARIMA model for the data before and after intervention respectively and see whether the estimated coefficients are quite different. The modelling procedure for intervention effect is as follows:

- 1 Use the longest data span to find a plausible ARIMA models
- 2 Estimate the various models over the entire sample period, including the effect of intervention
- 3 Perform diagnostic checks for the estimated coefficients (1, the estimated coefficients should be significant, 2, the residual should approximate white noise, 3, the tentative model should outperform plausible alternatives (AIC, SBC)).

Estimating the Effect of Metal Detector

Table 5.1 Metal Detectors and Skyjackings

	Pre-Intervention Mean	a_1	Impact Effect (c_0)	Long-Run Effect
Transnational (TS_t)	3.032 (5.96)	0.276 (2.51)	-1.29 (-2.21)	-1.78
US domestic (DS_t)	6.70 (12.02)		-5.62 (-8.73)	-5.62
Other skyjackings (OS_t)	6.80 (7.93)	0.237 (2.14)	-3.90 (-3.95)	-5.11

Notes:

¹t-Statistics are in parentheses.

²The long-run effect is calculated as $c_0/(1 - a_1)$.

ADLs and Transfer Functions

A natural extension of the intervention model is to allow $\{z_t\}$ sequence to be not deterministic. Consider following generalization

$$y_t = a_0 + A(L)y_{t-1} + C(L)z_t + B(L)\varepsilon_t \quad (3)$$

where $A(L)$, $C(L)$ and $B(L)$ are polynomials in the lag operator.

Remark 1: The model is called a **distributed lag** because it distributes the effects of z_t on y_t across several periods. The polynomial $C(L)$ is called the **transfer function** because it shows how a movement of exogenous variable z_t is transferred to the endogeneous y_t . The coefficients of $C(L)$, denoted by c_i are called the transfer function weights.

ADLs and Transfer Functions

Remark 2: Let $C(L) = c_0 + c_1L + \dots$. If $c_0 = 0$, the contemporaneous value of z_t does not affect y_t directly. In this case, $\{z_t\}$ is called a **leading indicator**.

Remark 3: Often we are not especially concerned about the moving average part and let $B(L) = 1$, so that (3) can be written as

$$y_t = a_0 + A(L)y_{t-1} + C(L)z_t + \varepsilon_t$$

which is called an **autoregressive distributed lag** model.

Properties of ADLs

Consider a special case of ADLs

$$y_t = a_1 y_{t-1} + c_d z_{t-d} + \varepsilon_t \quad (4)$$

where $\{z_t\}$ and $\{\varepsilon_t\}$ are white noise processes independent of each other. d is the delay or lag duration.

Define the **cross-correlation**

$$\rho_{yz}(i) = \frac{\text{cov}(y_t, z_{t-i})}{\sigma_y \sigma_z},$$

where σ_y and σ_z are respectively standard deviations of y_t and z_t . Plotting $\rho_{yz}(i)$ as a function of i yields the **cross-correlogram** or the cross-correlation function (CCF).

Properties of ADLs

Solving (4)

$$\begin{aligned}y_t &= c_d z_{t-d} / (1 - a_1 L) + \varepsilon_t / (1 - a_1 L) \\&= c_d [z_{t-d} + a_1 z_{t-d-1} + \cdots] + \varepsilon_t / (1 - a_1 L)\end{aligned}$$

To analyze the **cross-covariance**, the Yule-Walker procedure is adopted. We can obtain

$$\begin{aligned}E y_t z_t &= 0 \\E y_t z_{t-1} &= 0 \\&\dots \\E y_t z_{t-d} &= c_d \sigma_z^2 \\E y_t z_{t-d-1} &= c_d a_1 \sigma_z^2 \\&\dots\end{aligned}$$

Properties of ADLs

In summary, we have

$$E y_t z_{t-i} = 0 \text{ for all } i < d$$

$$E y_t z_{t-i} = c_d a_1^{i-d} \sigma_z^2 \text{ for all } i \geq d$$

Remark 1: The pattern can be generalized, suppose the model is

$$y_t = a_1 y_{t-1} + c_d z_{t-d} + c_{d+1} z_{t-d-1} + \varepsilon_t$$

Then we have

$$E y_t z_{t-i} = 0 \text{ for } i < d$$

$$= c_d \sigma_z^2 \text{ for } i = d$$

$$= (c_d a_1 + c_{d+1}) \sigma_z^2 \text{ for } i = d + 1$$

$$= a_1^{i-d-1} (c_d a_1 + c_{d+1}) \sigma_z^2 \text{ for } i > d + 1$$

Properties of ADLs

Remark 2: In summary, the shape of the cross-covariance function (CCVF) has the following characteristics

- 1 All the cross covariance will be zero until the first nonzero element of the polynomial $C(L)$
- 2 A spike in the CCVF indicates a nonzero elements of $C(L)$
- 3 The spikes decay (geometrically) at the rate a_1

Remark 3: For the second order process (typo in page 273), we have

$$\begin{aligned} E y_t z_{t-i} &= 0 \text{ for } i < d \\ &= c_d \sigma_z^2 \text{ for } i = d \\ &= c_d a_1 \sigma_z^2 \text{ for } i = d + 1 \\ &= a_1 E y_t z_{t-i+1} + a_2 E y_t z_{t-i+2} \text{ for } i \geq d + 2 \end{aligned}$$

Higher Order Input Process

In practice, $\{z_t\}$ is usually not a white noise process and the model now is

$$y_t = a_0 + A(L)y_{t-1} + C(L)z_t + \varepsilon_t \quad (5)$$

$$z_t = D(L)z_{t-1} + \varepsilon_{zt}, \quad (6)$$

where $D(L)$ is a polynomial of L , $\{\varepsilon_{zt}\}$ is a white noise.

Based on the equations (5) and (6), we can trace out three impulse response functions: (1) an ε_{zt} shock on the $\{z_t\}$ sequence; (2) an ε_t shock on the $\{y_t\}$ sequence and (3) an ε_{zt} shock on the $\{y_t\}$ sequence. To analyze the third one, we can substitute (6) into (5)

$$y_t = a_0 + A(L)y_{t-1} + \frac{C(L)}{1 - D(L)L} \varepsilon_{zt} + \varepsilon_t$$

Identification and Estimation

For the higher order input process, the identification and estimation of (6) is straightforward.

For (5), there is identification problem for $C(L)$. Consider a simple example.

$$\begin{aligned}y_t &= a_1 y_{t-1} + c_1 z_t + \varepsilon_t \\z_t &= d_1 z_{t-1} + \varepsilon_{zt}\end{aligned}$$

Substitution of z_t yields

$$y_t = a_1 y_{t-1} + c_1 (d_1 z_{t-1} + \varepsilon_{zt}) + \varepsilon_t$$

$\Rightarrow C(L)$ can be c_1 or $c_1 d_1 L$ (**identification problem**).

Identification and Estimation

We have two approaches to deal with this problem.

- 1 We do not care about the structure of $C(L)$ and we do estimation and model selection for the following model

$$y_t = a_0 + \sum_{i=1}^p a_i y_{t-i} + \sum_{i=0}^n c_i z_{t-i} + \varepsilon_t$$

- 2 We can filter y_t first, i.e. (Note the difference from the textbook p.275!),

$$\begin{aligned} (1 - D(L)L)y_t &= (1 - D(L)L)a_0 + (1 - D(L)L)A(L)y_{t-1} \\ &\quad + C(L)(1 - D(L)L)z_t + (1 - D(L)L)\varepsilon_t \end{aligned}$$

Define $y_{ft} = (1 - D(L)L)y_t$, we have

$$y_{ft} = (1 - D(L)L)a_0 + A(L)y_{ft-1} + C(L)\varepsilon_{zt} + (1 - D(L)L)\varepsilon_t$$

$C(L)$ can be identified in this model!

Identification and Estimation

The empirical procedure for the second approach.

- 1 Estimate the z_t sequence and obtain $\widehat{D}(L)$
- 2 Identify plausible candidates for $C(L)$. Construct the filter sequence and find plausible $C(L)$ via sample cross-correlogram.
- 3 Identify plausible candidates for the $A(L)$ function. Regress y_t on z_t and get the residuals whose sample PACF reveals information about $A(L)$
- 4 combine the results of step 2 and step 3 and estimate the full equation.

Limits to Structural Multivariate Estimation

Two important difficulties involved in fitting a multivariate equations such as a transfer function.

- 1 The first concerns the goal of fitting a parsimonious model
- 2 The second concerns the assumption of no feedback effect from $\{y_t\}$ sequence to $\{z_t\}$ sequence

Macroeconometric Modeling: Some Historical Background

Since Keynes, large scale econometric modeling becomes popular in macroeconomics.

Brookings Quarterly Econometric Model of the United States, as reported by Suits and Sparks (p. 208, 1965):

$$C_{NF} = 0.0656Y_D - 10.93(P_{CNF}/P_C)_{t-1} + 0.1889(N + N_{ML})_{t-1}$$

(0.0165) (2.49) (0.0522)

$$C_{NEF} = 4.2712 + 0.1691Y_D - 0.0743(ALQD_{HH}/P_C)_{t-1}$$

(0.0127) (0.0213)

where C_{NF} = personal consumption expenditures on food

Y_D = disposable personal income

P_{CNF} = implicit price deflator for personal consumption expenditures on food

P_C = implicit price deflator for personal consumption expenditures

N = civilian population

N_{ML} = military population including armed forces overseas

C_{NEF} = personal consumption expenditures for nondurables other than food

$ALQD_{HH}$ = end-of-quarter stock of liquid assets held by households

and standard errors are in parentheses.

Macroeconometric Modeling: Some Historical Background

Sims comments

"... what 'economic theory' tells us about them is mainly that any variable that appears on the right-hand-side of one of these equations belongs in principle on the right-hand-side of all of them. To the extent that models end up with very different sets of variables on the right-hand-side of these equations, they do so not by invoking economic theory, but (in the case of demand equations) by invoking an intuitive econometrician's version of psychological and sociological theory, since constraining utility functions is what is involved here. Furthermore, unless these sets of equations are considered as a system in the process of specification, the behavioral implications of the restrictions on all equations taken together may be less reasonable than the restrictions on any one equation taken by itself."

Macroeconometric Modeling: Some Historical Background

On the other hand, many monetarists use the **reduced-form** equation. E.g.

"St. Louis model" estimated by Anderson and Jordan (1968).

Using U.S. quarterly data from 1952 - 1968, they estimated the following reduced-form GNP determination equation:

$$\begin{aligned}\Delta Y_t = & 2.28 + 1.54\Delta M_t + 1.56\Delta M_{t-1} + 1.44\Delta M_{t-2} + 1.29\Delta M_{t-3} \\ & + 0.40\Delta E_t + 0.54\Delta E_{t-1} - 0.03\Delta E_{t-2} - 0.74\Delta E_{t-3} \quad (5.16)\end{aligned}$$

where $\Delta Y_t =$ change in nominal GNP

$\Delta M_t =$ change in the monetary base

$\Delta E_t =$ change in "high employment" budget deficit

Macroeconometric Modeling: Some Historical Background

Sims comments

Sims (1980) also points out several problems with this type of analysis. Sims's criticisms are easily understood by recognizing that (5.16) is a transfer function with two independent variables $\{M_t\}$ and $\{E_t\}$ and no lags of the dependent variable. As with any type of transfer function analysis, we must be concerned with two things:

1. *Ensuring that lag lengths are appropriate.* Serially correlated residuals in the presence of lagged dependent variables lead to biased coefficient estimates.
2. *Ensuring that there is no feedback between GNP and the money base or the budget deficit.* However, the assumption of no feedback is unreasonable if the monetary or fiscal authorities deliberately attempt to alter nominal GNP. As in the thermostat example, if the monetary authority attempts to control the economy by changing the money base, we cannot identify the “true” model. In the jargon of time-series econometrics, changes in GNP would “cause” changes in the money supply. One appropriate strategy would be to simultaneously estimate the GNP determination equation *and* the money supply feedback rule.

Macroeconometric Modeling: Some Historical Background

In summary

Comparing the two types of models, Sims (1980, pp. 14–15) states:

“Because existing large models contain too many incredible restrictions, empirical research aimed at testing competing macroeconomic theories too often proceeds in a single- or few-equation framework. For this reason alone, it appears worthwhile to investigate the possibility of building large models in a style which does not tend to accumulate restrictions so haphazardly. ... It should be feasible to estimate large-scale macromodels as unrestricted reduced forms, treating all variables as endogenous.”

⇒ **Vector autoregression (VAR).**

Introduction to VAR Analysis

Consider the bivariate system

$$y_t = b_{10} - b_{12}z_t + \gamma_{11}y_{t-1} + \gamma_{12}z_{t-1} + \varepsilon_{yt} \quad (7)$$

$$z_t = b_{20} - b_{21}y_t + \gamma_{21}y_{t-1} + \gamma_{22}z_{t-1} + \varepsilon_{zt} \quad (8)$$

where it is assumed that y_t and z_t are stationary, ε_{yt} and ε_{zt} are white noise processes with standard deviations σ_y and σ_z respectively and $\{\varepsilon_{yt}\}$ and $\{\varepsilon_{zt}\}$ are independent.

Equations (7) and (8) constitute a first order vector autoregression (VAR). However, the OLS estimation of the model will lead to inconsistency of the estimators since in (7), y_t is correlated with ε_{yt} and according to (8), z_t is correlated with ε_{yt} $\Rightarrow$ **endogeneity**.

Introduction to VAR Analysis

To solve the problem, we consider a transformed system. We have

$$\begin{bmatrix} 1 & b_{12} \\ b_{21} & 1 \end{bmatrix} \begin{bmatrix} y_t \\ z_t \end{bmatrix} = \begin{bmatrix} b_{10} \\ b_{20} \end{bmatrix} + \begin{bmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{21} & \gamma_{22} \end{bmatrix} \begin{bmatrix} y_{t-1} \\ z_{t-1} \end{bmatrix} + \begin{bmatrix} \varepsilon_{yt} \\ \varepsilon_{zt} \end{bmatrix}$$

and written as

$$Bx_t = \Gamma_0 + \Gamma_1 x_{t-1} + \varepsilon_t$$

Multiplication by B^{-1} allows us to obtain the VAR model in *standard form*

$$x_t = A_0 + A_1 x_{t-1} + e_t \tag{9}$$

where $A_0 = B^{-1}\Gamma_0$, $A_1 = B^{-1}\Gamma_1$ and $e_t = B^{-1}\varepsilon_t$.

Introduction to Analysis

Remark 1: We transform the system from a **structural VAR** to a **reduced form model**. And In the reduced form model, there are no contemporaneous terms on the right hand side of the equations $\implies$ no endogeneity.

Remark 2: $e_{1t} = (\varepsilon_{yt} - b_{12}\varepsilon_{zt})/(1 - b_{12}b_{21})$ and $e_{2t} = (\varepsilon_{zt} - b_{21}\varepsilon_{yt})/(1 - b_{12}b_{21})$. So that the new error term process are still white noise. However, the contemporaneous error covariance is now given by

$Ee_{1t}e_{2t} = E[(\varepsilon_{yt} - b_{12}\varepsilon_{zt})(\varepsilon_{zt} - b_{21}\varepsilon_{yt})]/(1 - b_{12}b_{21})^2 = -(b_{21}\sigma_y^2 + b_{12}\sigma_z^2)/(1 - b_{12}b_{21})^2$. We define the variance-covariance matrix as

$$\Sigma = E(e_t e_t') = \begin{bmatrix} \text{var}(e_{1t}) & \text{cov}(e_{1t}, e_{2t}) \\ \text{cov}(e_{1t}, e_{2t}) & \text{var}(e_{2t}) \end{bmatrix}.$$

Stability and Stationarity

Solving the reduced form model by iterating backward with n steps, we obtain

$$x_t = (I + A_1 + \cdots + A_1^n)A_0 + \sum_{i=0}^n A_1^i e_{t-i} + A_1^{n+1} x_{t-n-1}$$

Remark 1: It is clear that stability requires that the expression A_1^n vanishes as $n \rightarrow \infty$. Stability requires that the roots of $(1 - a_{11}L)(1 - a_{22}L) - a_{12}a_{21}L^2$ lie outside the unit circle. It corresponds to the requirement that the eigenvalues of the matrix A_1 lie inside the unit circle.

Remark 2: If we abstract from an initial condition, $\{x_t\}$ will be covariance stationary if the stability condition holds.

Stability and Stationarity

When stability condition holds

$$x_t = \mu + \sum_{i=0}^{\infty} A_1^i e_{t-i} \quad (10)$$

where $\mu = [I - A_1]^{-1} A_0$ is the unconditional mean of $\{x_t\}$ process.

For the variance-covariance matrix of $\{x_t\}$ (Note that the expressions in the textbook p.287 are not suitable)

$$E(x_t - \mu)(x_t - \mu)' = E\left[\sum_{i=0}^{\infty} A_1^i e_{t-i}\right]\left[\sum_{i=0}^{\infty} A_1^i e_{t-i}\right]'$$

and $\{e_t\}$ is a white noise vector with variance-covariance matrix Σ , thus

$$E(x_t - \mu)(x_t - \mu)' = (I - A_1)^{-1} \Sigma [(I - A_1)^{-1}]'$$

Estimation and Identification

Since the endogeneity problem does not appear in the standard VAR models, they can be estimated via OLS. And each equation in the systems can be estimated separately via OLS.

Sims's methodology entails little more than a determination of the appropriate variables to include in the VAR and a determination of the appropriate lag length. For a n -dimensional process with lag length p , the number of coefficients to be estimated is $n + pn^2$.

Unquestionable, a VAR will be over parameterized in that many coefficients will be insignificant. However, the goal is to find important interrelationships among the variables. Improperly imposing zero restrictions may waste important information.

Once the model is estimated, we can do forecasting.

Estimation and Identification

Our next question is: is it possible to recover the structural equation (7 and 8) from the standard form (9)?

The answer to this question is **No**, unless we are willing to appropriately restrict the primitive system (structural equation) $\implies$ estimating (9), we have 6 coefficient estimates and we calculate sample counterpart of $\text{var}(e_{1t})$, $\text{var}(e_{2t})$ and $\text{cov}(e_{1t}, e_{2t})$. However, the primitive system (7 and 8) contains 10 parameters (8 coefficients+2 standard deviation) $\implies$ the primitive system is unidentified.

Identification

Under identification:

$$x + y = 0$$

infinitely many solutions ($y = -x$) in linear algebra.

Exact identification:

$$x + y = 0$$

$$2x + y = 2$$

Solution $x = 2$, $y = -2$.

How about

$$x + y = 0$$

$$2x + 2y = 0$$

Identification

Over identification:

$$x + y = 0$$

$$2x + y = 2$$

$$3x + 4y = 4$$

No solutions. GMM.

Estimation and Identification

One way to identify the system proposed by Sims (1980) (recursive system): imposing restrictions on the primitive system such that $b_{21} = 0$.

Under this restriction

$$e_{1t} = \varepsilon_{yt} - b_{12}\varepsilon_{zt} \quad (11)$$

$$e_{2t} = \varepsilon_{zt} \quad (12)$$

so that

$$\text{var}(e_{1t}) = \sigma_y^2 + b_{12}^2\sigma_z^2$$

$$\text{var}(e_{2t}) = \sigma_z^2$$

$$\text{cov}(e_{1t}, e_{2t}) = -b_{12}\sigma_z^2$$

Three equations with three unknowns: $\sigma_z^2 \rightarrow b_{12} \rightarrow \sigma_y^2$.

Estimation and Identification

For the coefficients, we now have the matrix $B = \begin{bmatrix} 1 & b_{12} \\ 0 & 1 \end{bmatrix}$, so

that

$$\begin{bmatrix} y_t \\ z_t \end{bmatrix} = \begin{bmatrix} b_{10} - b_{12}b_{20} \\ b_{20} \end{bmatrix} + \begin{bmatrix} \gamma_{11} - b_{12}\gamma_{21} & \gamma_{12} - b_{12}\gamma_{22} \\ \gamma_{21} & \gamma_{22} \end{bmatrix} \begin{bmatrix} y_{t-1} \\ z_{t-1} \end{bmatrix} + \begin{bmatrix} \varepsilon_{yt} - b_{12}\varepsilon_{zt} \\ \varepsilon_{zt} \end{bmatrix} \quad (13)$$

Comparing (13) with (9), we have $a_{10} = b_{10} - b_{12}b_{20}$, $a_{11} = \gamma_{11} - b_{12}\gamma_{21}$, $a_{12} = \gamma_{12} - b_{12}\gamma_{22}$, $a_{20} = b_{20}$, $a_{21} = \gamma_{21}$ and $a_{22} = \gamma_{22}$. The coefficients in the primitive system can be solved accordingly. Also called **Choleski decomposition**.

The Impulse Response Function

The (stationary) VAR model can be written as vector moving average process (VMA). This allows us to trace to effects of some shocks on the variables. From (10) and (9)

$$x_t = \mu + \frac{1}{1 - b_{12}b_{21}} \sum_{i=0}^{\infty} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}^i \begin{bmatrix} 1 & -b_{12} \\ -b_{21} & 1 \end{bmatrix} \begin{bmatrix} \varepsilon_{yt-i} \\ \varepsilon_{zt-i} \end{bmatrix}$$

which will be rewritten as

$$x_t = \mu + \sum_{i=0}^{\infty} \phi(i) \varepsilon_{t-i} \quad (14)$$

Remark 1: The elements $\phi_{jk}(0)$, $j = 1, 2$, and $k = 1, 2$ are **impact multipliers** which represent the impact of a one-unit shock. The four sets of coefficients $\phi_{jk}(i)$ are called the **impulse response functions**.

The Impulse Response Function

Remark 2: As we have known, additional restrictions are required to identify the primitive system from the reduced form model. E.g., the recursive ordering (Choleski decomposition) required that $b_{21} = 0$ so that ε_{yt} shock has no direct (contemporaneous) effect on z_t . There is an **ordering** of the shocks.

Remark 3: There are not only one way of decomposition, e.g., we can let $b_{12} = 0$. In some instances, there might theoretical reasons for us to choose a decomposition (terrorism has no contemporaneous effects on tourism). Usually, there is no **priori** knowledge on how to choose a decomposition. And imposing a structure on the VAR system seems contrary to Sims' argument against incredible identification restrictions for the classic macroeconometric modeling.

Confidence Interval and Impulse Response

Consider the following estimate of an AR(1) process

$$y_t = \underset{(4.0)}{0.6} y_{t-1} + \widehat{\varepsilon}_t$$

Then we can form impulse response function $\phi(i) = 0.6^i$. Because the standard deviation for the coefficient is $0.6/4 = 0.15$, the confidence interval is (because of asymptotic normality **(not necessarily normality as in p.299)**)

$$[0.6 - 1.96 \times 0.15, 0.6 + 1.96 \times 0.15]$$

Confidence Interval and Impulse Response

Remark 1: Using the Delta method, we can construct confidence interval for $\phi(i)$, $i = 1, 2, \dots$. Suppose we have

$$\sqrt{T}(\widehat{\beta} - \beta) \rightarrow_d N(0, \sigma^2)$$

Then

$$\sqrt{T}[g(\widehat{\beta}) - g(\beta)] \rightarrow_d N(0, (g'(\beta))^2 \sigma^2)$$

In this case, $g(x)$ are x^i with $i = 1, 2, \dots$ (so that the decay pattern in the textbook p.299 is misleading)

Remark 2: When the process is an AR(p) process

$$y_t = a_0 + \sum_{i=1}^p a_i y_{t-i} + \varepsilon_t$$

this method is difficult to apply. A Monte Carlo **bootstrap** approach can be applied.

Confidence Interval and Impulse Response

The procedure for bootstrap confidence interval construction.

- 1 Estimate the model and get the coefficient estimators $\widehat{a}_i$, $i = 0, 1, 2, \dots, p$ and the residual sequence $\{\widehat{\varepsilon}_t\}$.
- 2 When the sample size is T , draw $\{\widehat{\varepsilon}_t\}$ with replacement and get the sequence $\{\widehat{\varepsilon}_t^s\}$. And then construct a new sequence

$$y_t^s = \widehat{a}_0 + \sum_{i=1}^p \widehat{a}_i y_{t-i}^s + \widehat{\varepsilon}_t^s$$

- 3 Estimate y_t^s as an AR(p) process. If the number of replications is B , you will construct the confidence intervals that are between the 2.5% quantile and 97.5% quantile of the replications.

The procedure for VAR process is almost the same except that the residuals is not one dimensional.

Variance Decomposition

Another useful tool in uncovering the interrelationships among the variables in the system in a forecast error decomposition. We have

$$E_t x_{t+1} = A_0 + A_1 x_t$$

and more generally

$$E_t x_{t+n} = (I + A_1 + \cdots + A_1^{n-1})A_0 + A_1^n x_t$$

So that the associated forecast error is

$$e_{t+n} + A_1 e_{t+n-1} + \cdots + A_1^{n-1} e_{t+1}$$

and because of (14), it is

$$\sum_{i=0}^n \phi_i e_{t+n-i}$$

Variance Decomposition

Thus for $\{y_t\}$ sequence, the forecast error is

$$\phi_{11}(0)\varepsilon_{yt+n} + \cdots + \phi_{11}(n-1)\varepsilon_{yt+1} + \phi_{12}(0)\varepsilon_{zt+n} + \cdots + \phi_{12}(n-1)\varepsilon_{zt+1}$$

Thus the forecast error variance is

$$\sigma_y(n)^2 = \sigma_y^2[\phi_{11}(0)^2 + \cdots + \phi_{11}(n-1)^2] + \sigma_z^2[\phi_{112}(0)^2 + \cdots + \phi_{11}(n-2)^2]$$

Remark: From the result, we can see the proportion of the movements in a sequence due to its own shocks and the the other variable. In the decomposition, the knowledge of the matrix B is necessary.

Testing Hypotheses

In a VAR model, we need to determine the number of variables n and the lag length p .

The likelihood ratio test statistics can be constructed for coefficient test. E.g., we want to test 8 variables against 12 variables, the likelihood statistic

$$T(\ln(|\Sigma_8|) - \ln(|\Sigma_{12}|))$$

The limiting distribution is χ^2 with degree of freedom equal to the number of restrictions.

We can use AIC and SBC to select the model

$$AIC = T\ln(|\Sigma|) + 2N$$

$$SBC = T\ln(|\Sigma|) + N\ln(T)$$

Granger Causality

A variable X Granger-cause Y if

$$P(Y(t+1) \in A \mid F_t) \neq P(Y(t+1) \in A \mid F_{-X,t})$$

where $F_{-X,t}$ denotes information up to time t excluding the information of X .

In a two equation VAR model with p lags, $\{y_t\}$ does not Granger cause $\{z_t\}$ if and only if all the coefficients in $A_{21}(L)$ are equal to 0. If all the variables in the VAR are stationary, the direct way to test Granger causality is to test

$$a_{21}(1) = \cdots = a_{21}(p) = 0$$

Structural VAR

Merits of Sims's (1980) VAR

- 1 All the variables are treated symmetrically so that all the variables are jointly endogeneous
- 2 The econometrician does not rely on any incredible identification restrictions
- 3 A good model for forecasting

Drawbacks

- 1 The beauty of this approach seems diminished when construction impulse response function
- 2 Being devoid of any economic content

⇒ Structural VAR.

Structural VAR

A simple way to identify the structural model is the Choleski decomposition (in the bivariate model, assuming $b_{21} = 0$). Forcing $b_{21} = 0$ is equivalent to assuming that an innovation in y_t does not have contemporaneous effect on $z_t \implies$ may lack theoretical foundation and the shocks are improperly identified. In Choleski decomposition, the additional restriction we impose is $n(n-1)/2$ where n is the number of the variables (put $n(n-1)/2$ terms equal to 0).

Sims (1986) and Bernanke (1986) propose modeling innovations using economic analysis. The basic idea is to estimate the relationships among the structural shocks using an economic model. **The important thing of identification is the number of restrictions rather than what restrictions we impose.**

参考文献

1. Antolín-Díaz, J. and Rubio-Ramírez, J., 2018. Narrative Sign Restrictions for SVARs. *American Economic Review*, 108. 2802-2829.
2. Caldara, D. and E., Herbst, 2019. Monetary Policy, Real Activity, and Credit Spreads: Evidence from Bayesian Proxy SVARs. *American Economic Journal: Macroeconomics*, 11, 157-192.
3. Ludvigson, S., S., Ma and S., Ng, 2021. Uncertainty and Business Cycles: Exogenous Impulse or Endogenous Response? *American Economic Journal: Macroeconomics*, forthcoming.
4. Milan, A., A., Sufi and V., Verner, 2017. Household Debt and Business Cycles Worldwide. *Quarterly Journal of Economics*, 132, 1755-1817.

参考文献

5. Brunnermeier, M., D., Palia, K., Sastry and C., Sims, 2021. Feedbacks: Financial Markets and Economic Activity. American Economic Review, Forthcoming.

6. Bernake, B.S., J., Bolvin and P., Eliazsz, 2005. Measuring the Effects of Monetary Policy: A Factor-Augmented Vector Autoregressive (FAVAR) Approach. Quarterly Journal of Economics, 120, 387-422.

7., Primiceri, G.E. (2005) Time Varying Structural Vector Autoregressions and Monetary Policy. Review of Economic Studies, 72, 821-852.

Summary and Conclusions

- Intervention analysis can be used to analyze the effect of a policy or action
- Transfer analysis is appropriate if the intervention sequence is stochastic
- VAR (proposed in Sims 1980) and structural VAR models are very important tools in econometrics

Homework

Exercise 4,6, 8

The homework must be submitted next week.

Time Series Analysis

Lecture 6: Models with Trend

Biqing Cai

Associate Professor

School of Economics, HUST

Spring, 2024

Outline of This Lecture

- 1 Introduction
- 2 Deterministic Trend Model
- 3 Stochastic Trend (Unit Root) Model
- 4 The Unit Root Tests
- 5 The Monte Carlo Method
- 6 Summary and Conclusions
- 7 Homework

Introduction

From chapter 3, we have learned several stylized facts in fact macroeconomics including the following two:

1. **Many of the series contain a clear trend.** Real and potential U.S. GDP, consumption and investment exhibit a decidedly upward trend (Figure 3.1).
3. **Shocks to a series can display a high degree of persistence.** Short term and long term interest rates (Figure 3.4).

They are the two objects to be studied in this chapter: deterministic trend and stochastic trend.

Real GDP

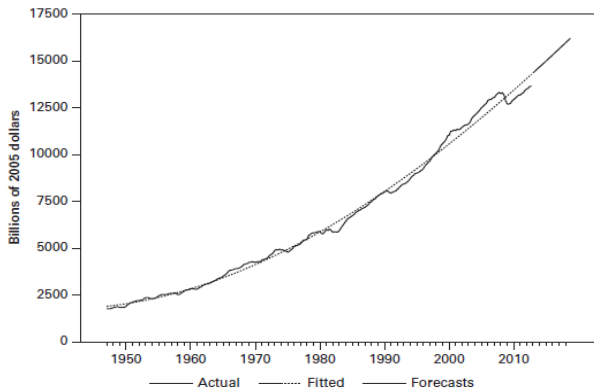


FIGURE 4.1 A Deterministic Trend in Real GDP

Interest Rates

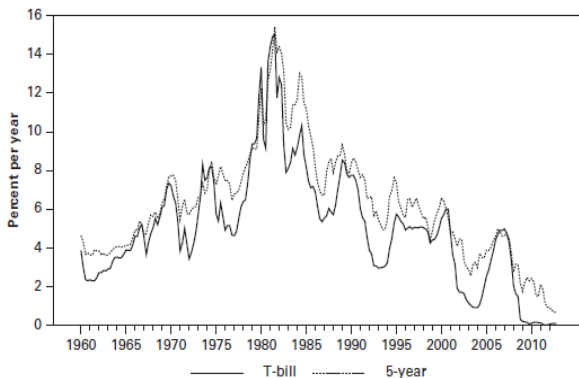


FIGURE 3.4 Short- and Long-Term Interest Rates

Introduction

Effects of nonstationarity

- The asymptotic theory is different from that of stationary time series
- The conventional tools are not usable (e.g., the Box-Jenkins methodology)

So that, we need to

- Transform the data to become stationary (detrend or difference)
- Using new asymptotic theory (Dickey-Fuller test, cointegration theory)

Deterministic Trend

In Figure 4.1, a cubic polynomial model is used to model the trend (Note the difference of the expression here and p.182, Eq(4.1))

$$\widehat{rgdp}_t = 1890.247 + \frac{9.108t}{(27.66)} + \frac{0.170t^2}{(4.09)9.108t} - \frac{0.0001t^3}{(8.70)(-2.07)}$$

The fitted values are shown as the dashed line in the figure.

Remark: The deterministic trend model assumes that there is a deterministic long-run growth rate of the economy.

Deterministic Trend

In most economic application, we encounter the following **trend stationary** model.

$$y_t = \alpha + \delta t + \varepsilon_t,$$

where $t = 1, \dots, T$ and $\{\varepsilon_t\}$ is a stationary process. We can use OLS to estimate the parameters and get

$$\begin{bmatrix} \widehat{\alpha} \\ \widehat{\delta} \end{bmatrix} = \left[\sum_{t=1}^T x_t x_t' \right]^{-1} \left[\sum_{t=1}^T x_t y_t \right]$$

where $x_t' = [1 \ t]$. And we can further show that the convergence rates of the coefficients are respectively $\sqrt{T}$ and $T^{3/2}$ with limiting normal distributions. I refer to Chapter 16 of Hamilton (1994) for more discussion.

Deterministic Trend

Once the model has been estimated, we can subtract the estimated values to get the residual sequence

$$\widehat{\varepsilon}_t = y_t - \widehat{\alpha} - \widehat{\delta}t$$

And then the Box-Jenkins methodology can be used to model the sequence $\{\widehat{\varepsilon}_t\}$.

Background of Unit Root

Traditional business cycle research decompose real macroeconomic variables into a long-run trend (secular) trend and a cyclical component such that a deviation from the trend is a wavelike motion called the **business cycle**.

The post-World War II experience suggests that the business cycles do not have a regular period. However, there is widespread belief that, over the long run, macroeconomic variables grow at a constant trend rate and any deviations from trend are eventually eliminated by the *invisible hand*.

Nelson and Plosser (1982) demonstrated that many important macroeconomic variables tend to be unit root process rather than trend stationary process. For example, the real GDP can be a difference stationary process rather than a trend stationary

Random Walk Model

Suppose we have

$$y_t = y_{t-1} + \varepsilon_t$$

where $\{\varepsilon_t\}$ is i.i.d. process with mean 0 variance σ^2 . Then the model is known as **random walk** model which is a special case of **unit root** model or **stochastic trend** model.

If y_0 is a given initial condition, we can obtain following solution

$$y_t = y_0 + \sum_{i=1}^t \varepsilon_i$$

Thus

$$E y_t = y_0$$

and

$$E_t y_{t+s} = y_t$$

Random Walk Model

Furthermore

$$\text{var}(y_t) = t\sigma^2$$

and

$$\text{cov}(y_t, y_{t-s}) = (t-s)\sigma^2$$

Thus, we can summarize that the process is not covariance stationary.

Random Walk Plus Drift Model

The random walk plus drift model is

$$y_t = y_{t-1} + a_0 + \varepsilon_t$$

Thus given initial condition y_0 , we have

$$y_t = y_0 + a_0 t + \sum_{i=1}^t \varepsilon_i$$

Thus there are two nonstationary components: a linear deterministic trend and the stochastic trend $\sum_{i=1}^t \varepsilon_i$.

Differencing

For the unit root process, we can difference the process to get a stationary process. Consider the random walk plus drift model, we have

$$\Delta y_t = a_0 + \varepsilon_t$$

It is easy to check it is a stationary process.

For a general ARIMA(p,1,q) model

$$A(L)y_t = B(L)\varepsilon_t$$

Suppose $A(L)$ has a single unit root and that $B(L)$ has all roots outside the unit circle, we have

$$(1 - L)A^*(L)y_t = B(L)\varepsilon_t$$

Differencing

This suggests

$$A^*(L)\Delta y_t = B(L)\varepsilon_t$$

and all roots of $A^*(L)$ are outside the unit circle $\implies \Delta y_t$ is a stationary process.

Remark: We can generalize the process to ARIMA(p,d,q) model. The general point is that the d-th difference of a process with d unit roots is a stationary process. Such a process is integrated of order d and is denoted by I(d). The unit root process is the I(1) process.

The Unit Root Tests

I will use the materials in Phillips and Xiao (1998) instead.

The Monte Carlo Method

In modern econometrics, Monte Carlo simulation plays an important role in

- Verifying the theoretical results
- Examining the finite sample performance of the estimators or test statistics
- Detecting possible problems of the methods

The Monte Carlo bootstrap method discussed in Lecture 5 for confidence interval construction is an example.

Spurious Regression

Possibly the most famous Monte Carlo simulation in the history of econometrics is the **spurious regression** in Granger and Newbold (1974). They consider regression

$$y_t = a_0 + a_1 z_t + e_t$$

where

$$y_t = y_{t-1} + \varepsilon_{yt}$$

$$z_t = z_{t-1} + \varepsilon_{zt}$$

where $\{\varepsilon_{yt}\}$ and $\{\varepsilon_{zt}\}$ are i.i.d. processes independent of each other.

Since the two processes are independent, we should expect that we can not reject the hypothesis that $a_1 = 0$.

Spurious Regression

However,

- At the 5% significance level, they were able to reject the null hypothesis $a_1 = 0$ in approximately 75% of the cases (replications)
- The regression usually had very high R^2 values and the estimated residuals exhibits a high degree of autocorrelations

This is the so-called spurious regression phenomenon.

Remark: Phillips (1987) provides a theoretical explanation for the results. And when $y_t = \mu_t + \varepsilon_{yt}$, $z_t = \mu_t + \varepsilon_{zt}$ where μ_t is a unit root, ε_{yt} and ε_{zt} are stationary, they are **cointegrated**.

The Monte Carlo Method

If we want to do the unit root testing without relying on the asymptotic theory (the t -statistics does not follow asymptotic normal distribution but **Dickey-Fuller distribution**), Monte Carlo method is a good alternative. The procedure for generating Dickey-Fuller distribution (with sample size 100) is as follows:

- 1 Simulate 100 $\{\varepsilon_t\}$ standard normal variable
- 2 We will generate $y_t = y_{t-1} + \varepsilon_t$ and an initial value y_0 is required (to eliminate the effect of initial value)
- 3 We estimate $\Delta y_t = a_0 + \gamma y_{t-1} + \varepsilon_t$ and obtain the t -statistic for the null $\gamma = 0$.
- 4 Repeat steps 1-3, for N times. We can get the Dickey-Fuller distribution based on the empirical distribution.

The Empirical Dickey-Fuller Distribution

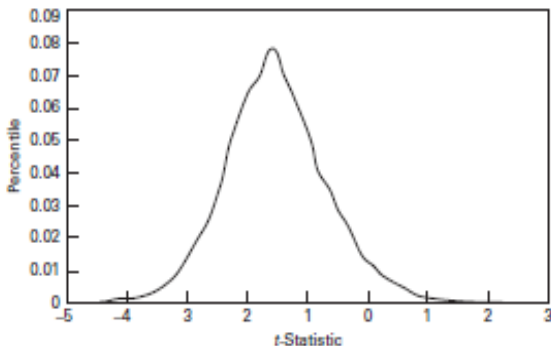


FIGURE 4.6 The Dickey-Fuller Distribution

Summary and Conclusions

- This Chapter discusses deterministic trend and stochastic trend models
- Introduces unit root testing problem (DF test, ADF test, KPSS test)
- Explains how Monte Carlo simulations can be used to derive critical values of the tests

Homework

Exercise 2, 3

The homework must be submitted one week later.

Time Series Analysis

Lecture 7: Cointegration and Error Correction Models

Biqing Cai

Associate Professor

School of Economics, HUST

Spring, 2024

Outline of This Lecture

- 1 Introduction
- 2 Cointegration Analysis
- 3 Testing Cointegration
- 4 Summary and Conclusions
- 5 Homework

Introduction

In Lecture 6, we have mentioned

- Unit root is a stylized fact in macroeconomic data
- Inappropriate modeling of unit root process can lead to spurious regression
 - ⇒ We should be careful in modeling unit root processes
 - ⇒ Cointegration models are the tool to analyze the relationship of unit root processes.

Introduction

According to the conventional Box-Jenkins methodology, the unit root processes will be differenced before modeling

However, Differencing leads to loss of information

It is possible that there are linear combinations of integrated processes that is stationary. Such variables are said to be **cointegrated**

In the presence of cointegrated variables, it is possible to model the long run (cointegration) and short run dynamics simultaneously

Introduction

Examples of cointegration relationships in Economics

- Money demand studies. $m_t = \beta_0 + \beta_1 p_t + \beta_2 y_t + \beta_3 r_t + e_t$ where m_t denotes demand for money, p_t denotes price level, y_t denotes real income, r_t denotes interest rate. $\{e_t\}$ is a stationary sequence.
- Consumption function theory. According to permanent income hypothesis, $c_t = \beta y_t^p + c_t^t$, where y_t^p is the permanent income and the transitory income c_t^t is stationary.
- Unbiased forward rate hypothesis. $s_{t+1} = f_t + \varepsilon_{t+1}$ where s_{t+1} is the $t + 1$ period exchange rate and f_t is the t period one-period forward rate.
- Purchasing power parity. $e_t + p_t^* - p_t$ is required to be stationary, where e_t denotes log foreign exchange rate, p_t and p_t^* denote the logs of domestic and foreign price level.

General Framework

All the examples illustrate the concept of **cointegration** as in Engle and Granger (1987) with

$$e_t = \beta' x_t \quad (1)$$

where $\beta = (\beta_1, \dots, \beta_n)'$, $x_t = (x_{1t}, \dots, x_{nt})'$ (**note the difference with p.346**) is a vector of $I(1)$ processes and e_t is a stationary process called the **equilibrium error**.

According to Engle and Granger (1987), the components of x_t are said to be **cointegrated of order d , b** , denoted by $x_t \sim CI(d, b)$ if

- 1 All components of x_t are integrated of order d
- 2 There exists a vector β such that $\beta' x_t$ is integrated of order $(d - b)$ with $b > 0$. β is called the **cointegrating vector**

General Framework

Remark 1: The most widely used model in econometrics is the $CI(1, 1)$ model.

Remark 2: Cointegration typically refers to a **linear** combination of nonstationary variables. The cointegrating vector is not unique: if β is a cointegrating vector, so is $\lambda\beta$ where $\lambda \neq 0$. Typically, one of the coefficient is normalized to 1. And there is possibility of nonlinear cointegration.

General Framework

Remark 3: There may be more than one independent cointegrating vectors. Suppose $x_{2t} = x_{1t} + e_{1t}$ and $x_{3t} = 2x_{1t} + e_{2t}$, the cointegrating vector can be $(1, -1, 0)'$ or $(2, 0, -1)'$. It will become more clear when we mention common stochastic trend later.

Remark 4: Multicointegration. The definition of Engle and Granger (1987) requires all the components of x_t have the same integrating order. Lee and Granger (1990) introduce multicointegration referring to the case where x_{1t} and x_{2t} are $I(2)$ and there exists a linear combination of the two variables to be $I(1)$, furthermore, this combination of x_{1t} and x_{2t} is cointegrated with other $I(1)$ variables.

Cointegration and Common Stochastic Trend

Stock and Watson (1988): cointegrating variables share common stochastic trends.

$$y_t = \mu_{yt} + e_{yt}$$

$$z_t = \mu_{zt} + e_{zt}$$

where μ_{it} is a random walk process representing stochastic trend in variable i , e_{it} is a stationary component of variable i .

When y_t and z_t are $CI(1, 1)$, we should have there exists $(\beta_1, \beta_2)'$ such that $\beta_1 y_t + \beta_2 z_t$ is stationary, so that

$$\beta_1 \mu_{yt} + \beta_2 \mu_{zt} = 0$$

which implies

$$\mu_{yt} = -\beta_2 \mu_{zt} / \beta_1$$

Cointegration and Common Stochastic Trend

Remark 1: Thus, up to the scalar $-\beta_2/\beta_1$, the two $I(1)$ processes have the same stochastic trend if they are cointegrated of order $(1,1)$.

Remark 2: The essential insight of Stock and Watson (1988) is that the parameters of the cointegrating vector must be such that they purge the trend from the linear combination. The cointegrating vector (in the bivariate case) is unique up to a normalizing scalar such that $\beta_3 y_t + \beta_4 z_t$ cannot be stationary unless $\beta_3/\beta_4 = \beta_1/\beta_2$.

Remark 3: When the dimensional is larger than 2, cointegration will occur whenever the trend in one variable can be express as a linear combination of the trends in other variables.

Cointegration and Error Correction

A principle of cointegrated variables is that their paths are influenced by the extent of any deviation from long run equilibrium. The variables will respond to the disequilibrium to return to long-run equilibrium.

The economic theory suggests a long run equilibrium relationship between the two processes. If the gap between the long term and short term interest rates is large relative to long run equilibrium, the short term interest rate will rise relative to long term interest rate.

The discussion leads to **error correlation** model

$$\Delta r_{St} = \alpha_S (r_{Lt-1} - \beta r_{St-1}) + \varepsilon_{St} \quad \alpha_S > 0$$

$$\Delta r_{Lt} = -\alpha_L (r_{Lt-1} - \beta r_{St-1}) + \varepsilon_{Lt} \quad \alpha_L > 0$$

Cointegration and Error Correction

Remark 1: As specified, the short term and long term interest rates change in response to stochastic shocks (ε_{St} , ε_{Lt}) and in previous period's deviation from the long run equilibrium. If the deviation is positive, the short term interest rate will rise and the long term interest rate would fall.

Remark 2: We can see that relationship between error correction models and cointegrated variables. Since Δr_{St} is stationary $I(0)$ process, it follows that $r_{Lt-1} - \beta r_{St-1}$ must be stationary, which suggests the cointegrating vector is $(1, -\beta)'$.

Remark 3: α_S and α_L have the interpretation of **speed of adjustment** parameters. The larger α_S is, the greater the response of r_{St} to deviation from long run equilibrium.

Cointegration and Error Correction

The result can be extended to n -variable model.

$$\Delta x_t = \pi_0 + \pi x_{t-1} + \sum_{i=1}^p \pi_i \Delta x_{t-i} + \varepsilon_t$$

where x_t is an $n \times 1$ vector of $I(1)$ variables, π_0 is an $n \times 1$ vector of intercept, π_i are $n \times n$ coefficient matrices, π is matrix with a least one element not equal to 0, ε_t is a vector of stationary error terms.

The equation can be rewritten as

$$\pi x_{t-1} = \Delta x_t - \pi_0 - \sum_i \pi_i \Delta x_{t-i} - \varepsilon_t$$

Since the right hand side (RHS) variables are stationary, so are the left hand side (LHS) variables. Thus each row of π is a cointegrating vector of x_t .

Cointegration and Error Correction

Compared with stationary VAR model, the key feature is the presence of matrix π .

- 1 If all the elements in π are zero, then it becomes traditional VAR model in first difference. In such case, there is no error correction representation since Δx_t does not respond to previous period's deviation from the long run equilibrium
- 2 If one or more elements of π differs from zero, Δx_t respond to deviation from long run equilibrium. Hence, estimating x_t in a VAR in first difference is inappropriate if x_t has an error correction representation. The omission of πx_{t-1} leads to misspecification error.

Cointegration and Error Correction

To examine the relationship of cointegration and error correction, consider the bivariate VAR model

$$y_t = a_{11}y_{t-1} + a_{12}z_{t-1} + \varepsilon_{yt}$$

$$z_t = a_{21}y_{t-1} + a_{22}z_{t-1} + \varepsilon_{zt}$$

Thus we can derive

$$y_t = \frac{(1 - a_{22}L)\varepsilon_{yt} + a_{12}L\varepsilon_{zt}}{(1 - a_{11}L)(1 - a_{22}L) - a_{12}a_{21}L^2}$$

$$z_t = \frac{a_{21}L\varepsilon_{yt} + (1 - a_{11})L\varepsilon_{zt}}{(1 - a_{11}L)(1 - a_{22}L) - a_{12}a_{21}L^2}$$

Cointegration and Error Correction

Define $\lambda = 1/L$ and write the characteristic equation as

$$\lambda^2 - (a_{11} + a_{22})\lambda + (a_{11}a_{22} - a_{12}a_{21}) = 0$$

- 1 If both λ_1 and λ_2 lie inside the unit circle, the variables are stationary, so that they are not CI(1,1).
- 2 If either root lie inside the unit circle, the solutions are explosive. They are not CI(1,1).
- 3 If $a_{12} = a_{21} = 0$, y_t then z_t will be unit roots, if $a_{11} = a_{22} = 1$. $\lambda_1 = \lambda_2 = 1$, they are not cointegrated.
- 4 For y_t and z_t to be CI(1,1), it is necessary that one characteristic root to be 1 and the other with absolute value less than 1.

Cointegration and Error Correction

In the CI(1,1) case, when $\lambda_1 = 1$, we have

$$y_t = \frac{(1 - a_{22}L)\varepsilon_{yt} + a_{12}L\varepsilon_{zt}}{(1 - L)(1 - \lambda_2 L)}$$

and

$$\Delta y_t = \frac{(1 - a_{22}L)\varepsilon_{yt} + a_{12}L\varepsilon_{zt}}{(1 - \lambda_2 L)}$$

which is stationary because $|\lambda_2| < 1$.

Because the roots of the characteristic equation are

$$\frac{(a_{11} + a_{22}) \pm \sqrt{(a_{11}^2 + a_{22}^2) - 2a_{11}a_{22} + 4a_{12}a_{21}}}{2}$$

Cointegration and Error Correction

Because the larger root is 1, we have

$$a_{11} = [(1 - a_{22}) - a_{12}a_{21}]/(1 - a_{22})$$

and because $|\lambda_2| < 1$, we have

$$a_{22} > -1$$

and

$$a_{12}a_{21} + a_{22}^2 < 1$$

Rewrite the VAR model as

$$\Delta y_t = (a_{11} - 1)y_{t-1} + a_{12}z_{t-1} + \varepsilon_{yt}$$

$$\Delta z_t = a_{21}y_{t-1} + (a_{22} - 1)z_{t-1} + \varepsilon_{zt}$$

Cointegration and Error Correction

And using the restriction implied by $CI(1,1)$, we have

$$\Delta y_t = -[a_{12}a_{21}/(1 - a_{22})]y_{t-1} + a_{12}z_{t-1} + \varepsilon_{yt}$$

$$\Delta z_t = a_{21}y_{t-1} - (1 - a_{22})z_{t-1} + \varepsilon_{zt}$$

which further implies

$$\Delta y_t = a_y(y_{t-1} - \beta z_{t-1}) + \varepsilon_{yt}$$

$$\Delta z_t = a_z(y_{t-1} - \beta z_{t-1}) + \varepsilon_{zt}$$

with $a_y = -a_{12}a_{21}/(1 - a_{22})$, $\beta = (1 - a_{22})/a_{21}$ and $a_z = a_{21}$.

This is the **error correction** model.

Cointegration and Error Correction

Remark 1: The restrictions necessary to ensure that the variables are $CI(1,1)$ guarantee the error correction model exists. **Granger representation theorem** states for any set of $I(1)$ variables, error correction and cointegration are equivalent representations.

Remark 2: A cointegration necessitates coefficient restrictions in a VAR model. In the model

$$\Delta x_t = \pi x_{t-1} + \varepsilon_t$$

it is inappropriate to estimate VAR cointegration model using only first differences. The insight of Johansen (1988) and Stock and Watson (1988) is to use the rank of π to determine whether the variables are cointegrated.

Cointegration and Error Correction

Remark 3: Cointegration requires that a_{12} and a_{21} cannot be both 0. However, it is possible that one of a_{12} and a_{21} is zero. If $a_{12} = 0$, then y_t responds only to its disturbance, and it is said to be **weakly exogenous**. It is necessary to reinterpret Granger causality in a cointegrated system. $\{y_t\}$ does not Granger cause $\{z_t\}$ if lagged values Δy_{t-i} does not enter the Δz_t equation and if z_t does not respond to long run deviation. When $a_{12} = 0$, $\{y_t\}$ is weakly exogenous and is not Granger caused by $\{z_t\}$.

Remark 4: In the n -variables case, if the rank of π is equal to $r < n$, there are r cointegrating vectors. This fact is closely related to Johansen test.

Engle-Granger Test

Engle and Granger (1987) propose following procedure to determine whether some variables are $CI(1,1)$.

- 1 Pretest the variables for their order of integration (ADF test, PP test)
- 2 If the variables are $I(1)$, estimate the long run equilibrium relationship

$$y_t = \beta_0 + \sum_{i=1}^n \beta_i x_{it} + e_t$$

and get the residual sequence $\{\widehat{e}_t\}$. Then do the unit root test on the residual to test **no cointegration** against cointegration.

- 3 If the variables are cointegrated, estimate the error correction model replacing the long run deviation with the residual
- 4 Assess the model adequacy

Engle-Granger Test

Remark 1: Steps 1 and 2 are called Engle-Granger (EG) two-step procedure in the literature.

Remark 2: In the second step, the estimator is super consistent with rate T . However, the limiting distributions are different from those in stationary time series. The Phillips and Hansen (1990) procedure can be used to construct asymptotically valid test.

Remark 3: Section 6 of the textbook provides analysis with exchange rate and price levels using EG methodology.

Remark 4: In the EG procedure, we need to treat the variables on different status (explanatory and explained variables); the second step inference may relies on the first step estimation (the first step cointegrating vector may not be unique)

Johansen Test

The Johansen (1988) procedure relies heavily on the relationships between the rank of matrix π and its characteristic roots. It can circumvent the use of two-step estimators and can estimate and test for the presence of multiple cointegrating vectors.

Consider an n -variable system

$$x_t = Ax_{t-1} + \varepsilon_t$$

So that

$$\Delta x_t = (A - I)x_{t-1} + \varepsilon_t \equiv \pi x_{t-1} + \varepsilon_t$$

The rank of π equals to the number of cointegrating vectors (and the number of common stochastic trends is $n - \text{rank}(\pi)$).

- When $\text{rank}(\pi)=0$, there is no linear combination to make a stationary process
- When $\text{rank}(\pi)=n$, all the variables are stationary

Johansen Test

As with the ADF test, the model can be generalized to higher autoregressive processes

$$x_t = A_1 x_{t-1} + \cdots + A_p x_{t-p} + \varepsilon_t$$

We can add and subtract $A_p x_{t-p+1}$ on the equation and have

$$x_t = A_1 x_{t-1} + \cdots + A_{p-2} x_{t-p-2} + (A_{p-1} + A_p) x_{t-p+1} - A_p \Delta x_{t-p+1} + \varepsilon_t$$

Repeating the procedure, we can derive

$$\Delta x_t = \pi x_{t-1} + \sum_{i=1}^{p-1} \pi_i \Delta x_{t-i} + \varepsilon_t$$

with $\pi = -(I - \sum_{i=1}^p A_i)$ and $\pi_i = -\sum_{j=i+1}^p A_j$.

Johansen Test

Clearly, the long run properties systems are described by the properties of the matrix π . There are three cases of interest:

- 1 Rank $(\pi) = 0$; the system is non-stationary, with no cointegration between the variables considered; this is the only case in which non-stationarity is correctly removed simply by taking the first differences of the variables;
- 2 Rank $(\pi) = n$, full rank; the system is stationary;
- 3 Rank $(\pi) = r$, $1 \leq r < n$; the system is non-stationary but there are r cointegrating relationships among the n variables (thus $n - r$ independent common stochastic trends). In this case $\pi = \alpha\beta'$, where α is an $(n \times r)$ matrix of weights and β is an $(n \times r)$ matrix of parameters determining the cointegrating relationships.

Johansen Test

The rank of π is crucial in determining the number of cointegrating vectors. Johansen's procedure is therefore based on the fact that the rank of a matrix equals the number of its characteristic roots that differ from zero.

we associate to them estimates for the n characteristic roots and order them as follows: $\lambda_1 > \lambda_2 > \dots > \lambda_n$ according to Johansen's procedure (note the the expression in the textbook is not correct. The eigenvalues used in the Johansen test are not eigenvalues of π . They are positive or zero while the eigenvalues of π are zero or negative). If the variables are not cointegrated, then the rank of π is 0, and all the λ_i are 0.

If, $\text{rank}(\pi) = 1$ and $0 < \lambda_1 < 1$ then $\ln(1 - \lambda_1)$ is negative and $\ln(1 - \lambda_2) = \dots = \ln(1 - \lambda_n) = 0$.

Johansen Test

In practice, we do not have the true values of λ_i , but we can get their estimators.

The test for the number of characteristic roots that are significantly different from 0 (**not 1 as in the textbook p.378**) can be constructed as

$$\lambda_{trace}(r) = -T \sum_{i=r+1}^n \ln(1 - \widehat{\lambda}_i)$$
$$\lambda_{max}(r, r+1) = -T \ln(1 - \widehat{\lambda}_{r+1})$$

where $\widehat{\lambda}_i$ are the estimated eigenvalues and T is the number of observations.

Johansen Test

Remark 1: The H_0 of the first statistic is that the number of **distinct** cointegrating vectors is less than or equal to r , and the H_1 is that the number is larger than r ; The H_0 of the second statistic is that the number of cointegrating vector is r and the H_1 is that the number is $r + 1$.

Remark 2: The results of the two test statistics can conflict. The λ_{max} test has the sharper alternative and is usually preferred.

Summary and Conclusions

- Many economic theories imply a linear combination of nonstationary variables must be stationary. These variables are called cointegrated variables.
- Cointegration is closely related two other important concepts: common stochastic trend and error correction
- EG procedure for testing cointegration and Johansen test

Homework

Exercise 1, 6

The homework must be submitted at the last class.